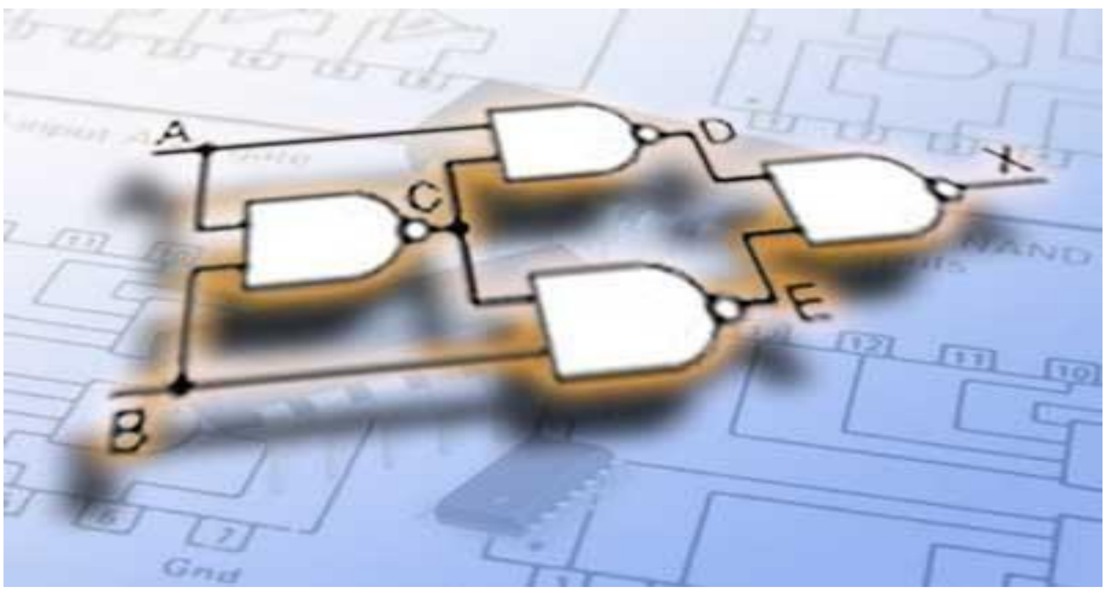
Department of Electrical, Electronics and Communication Engineering

INTRODUCTION TO DIGITAL SYSTEM (IDS)

Lab Manual

[G2UC101B]



B. Tech 1st Year

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About Laboratory Manual

This manual is intended for the First-year students of engineering branches in the subject "Introduction to Digital System". This manual typically contains practical/Lab Sessions related Digital Electronics covering various aspects related to the subject to enhance understanding. Students are advised to thoroughly go through this manual rather than only topics mentioned in the syllabus. As practical aspects are the key to understanding and conceptual visualization of theoretical aspects covered in the books.

Course Code: G2UC101B

Title of the course: Introduction to Digital System Lab

1. Dos and Don'ts in the laboratory
2. Lab Experiments:

To understand the practicability of Analog and Digital Electronics, the list of **Learning Outcomes** based on experiments to be performed (at least 10) in the laboratory is given below:

By the end of Lab. Session, the students should be able to:

1. Identify common TTL IC numbers (e.g., (IC-7408), OR (IC-7432), NAND (IC-7400), NOR (IC-7402), NOT (IC-7404) and XOR (IC-7486), etc.) and learn how to correctly connect them in a circuit for verification of respective Truth Tables

2. Demonstrate knowledge of DeMorgan's laws and their significance in Boolean function

3. Derive the implementation of basic gates (NOT, AND, OR), NOR, XOR and XNOR using only NAND and NOR gates

4. Design Half Adder and Full Adder circuits using basic logic gates (AND, OR, XOR) and differentiate between Half Adder and Full Adder in terms of inputs, outputs, and functionality

5. Derive Boolean expressions for *Difference* and *Borrow* outputs of Half and Full Subtractors and Differentiate between Half and Full Subtractor in terms of inputs, outputs, and borrow concept

6. Identify pin configuration and function of the IC-74151 (MUX) and realize how a multiplexer can implement combinational logic circuits

7. Explain the working principle of a De-Multiplexer (De-MUX) as a data distributor and understand the role of select lines, enable pins, input, and outputs in IC-74138 (3-to-8 line De-MUX/decoder)

8. Realize the construction of the equivalent design of S-R flip-flop using NOR/NAND gates as memory elements in sequential circuits Design and test of an S-R flip-flop using NOR/NAND gates/7400

9. Apply various input combinations to a J-K flip-flop (7476) and observe corresponding output (Q, Q') experimentally

10. Use various input combinations to verify truth table of a D and T flip-flop (JK-7476, NOT-7404, D-7474)

11. Construct a decoder (n input lines into 2^n output lines) and encoder (2^n input lines into n output lines) using only basic gates (AND, OR, NOT)

12. Derive and design the logic circuit for a Mod-8 Up-Down counter (Ripple) using JK or T flip-flops

13. Learn how to design synchronous counters (Mod-3 and Mod-2) using flip-flops (typically JK)Realization of Synchronous MOD-3 and MOD-2 Counters

Experiments beyond the curriculum

14. Gain hands-on experience in programming Arduino using C/C++ or Arduino IDE in the form of Mini projects using programmable ARDUINO

Project 1: Controlling the Light Emitting Diode (LED) with a push button.
Project 2: Interfacing of temperature sensor LM35 with Arduino

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DO'S and DON'TS in Laboratory

1. Come fully prepared for the experiment in the laboratory.
2. Check for appropriate power supply before connecting to the equipment.
3. Decide the appropriate range of the measuring instruments on the basis of quantity to be measured.
4. Make the connections without connecting the leads to the supply.
5. Re-check the connections and show it to the teacher /instructor before switching-on the power supply to the circuit.
6. Energize the circuit only with the permission of the teacher/instructor.
7. After the experiment, disconnect the connections and put back the connecting wires/leads at appropriate place.
8. Return all the apparatus to the lab-staff.
9. In case of shock, switch-off the power supply immediately.
10. Strictly follow the procedure given with the respective experiments.
11. Avoid loose connections.
12. Don't touch the main power supply leads with bare hand and avoid body earth.
13. Don't use the mobile phones during laboratory.

EXPERIMENT – 1

LEARNING OUTCOMES: -

By the end of Lab. Session, the students should be able to identify common TTL IC numbers (e.g., (IC-7408), OR (IC-7432), NAND (IC-7400), NOR (IC-7402), NOT (IC-7404) and XOR (IC-7486), etc.) and learn how to correctly connect them in a circuit for verification of respective Truth Tables.

APPARATUS REQUIRED: -

Digital lab kit, single strand wires, breadboard, TTL IC's AND (IC-7408), OR (IC-7432), NAND (IC-7400), NOR (IC-7402), NOT (IC-7404) and XOR (IC-7486).

THEORY: -

Logic gates are idealized or physical devices implementing a Boolean function, which it performs a logical operation on one or more logical inputs and produce a single output. Depending on the context, the term may refer to an ideal logic gate, one that has for instance zero rise time and unlimited fan out or it may refer to anon-ideal physical device. The main hierarchy is as follows: - 1.
Basic Gates
2. Universal Gates
3. Advanced Gates

BASIC GATES

AND GATE:

The AND gate performs a logical multiplication commonly known as AND function. The output is high when both the inputs are high. The output is low level when any one of the inputs is low.

OR GATE:

The OR gate performs a logical addition commonly known as OR function. The output is high when any one of the inputs is high. The output is low level when both the inputs are low.

NOT GATE:

The NOT gate is called an inverter. The output is high when the input is low. The output is low when the input is high.

UNIVERSAL GATES

NAND GATE:

The NAND gate is a contraction of AND-NOT. The output is high when both inputs are low and any one of the input is low. The output is low level when both inputs are high.

NOR GATE:

The NOR gate is a contraction of OR-NOT. The output is high when both inputs are low. The output is low when one or both inputs are high.

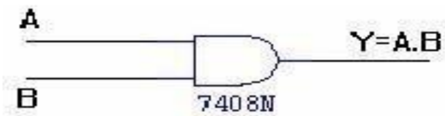
ADVANCED GATES

X-OR GATE:

The output is high when any one of the inputs is high. The output is low when both the inputs are low and both the inputs are high.

AND GATE:

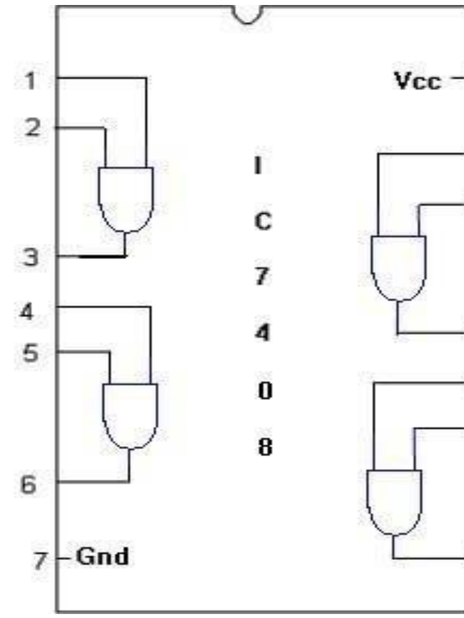
SYMBOL:



TRUTH TABLE

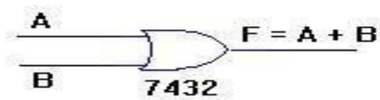
A	B	A.B
0	0	0
0	1	0
1	0	0
1	1	1

PIN DIAGRAM:



OR GATE:

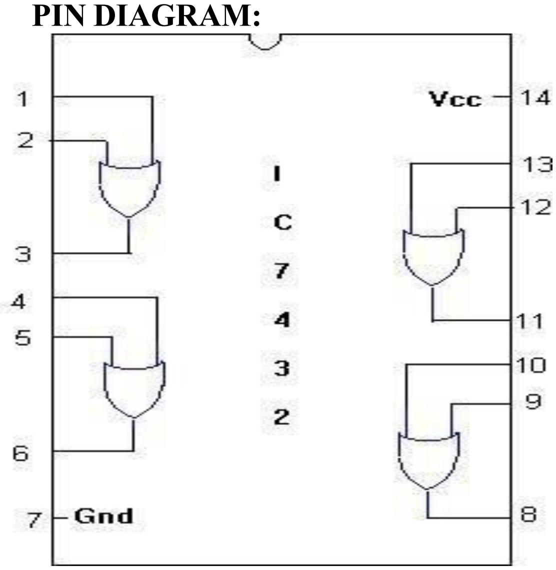
SYMBOL:



TRUTH TABLE

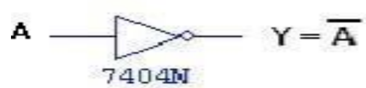
A	B	A+B
0	0	0
0	1	1
1	0	1
1	1	1

PIN DIAGRAM:

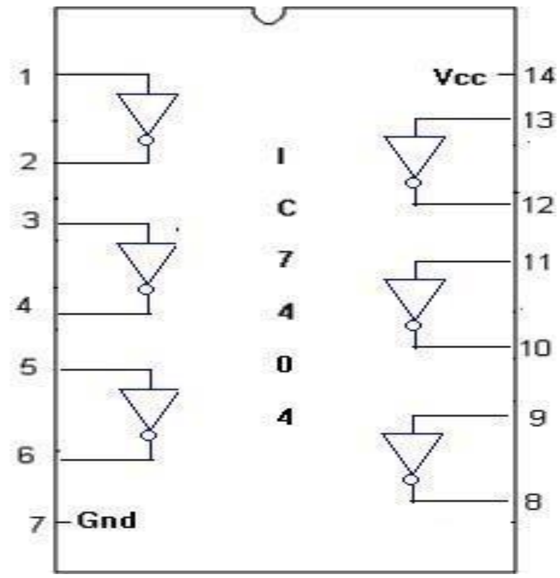


NOT GATE:

SYMBOL:



PIN DIAGRAM:

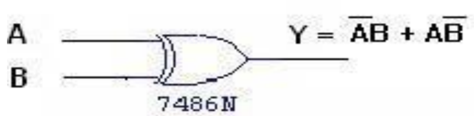


TRUTH TABLE :

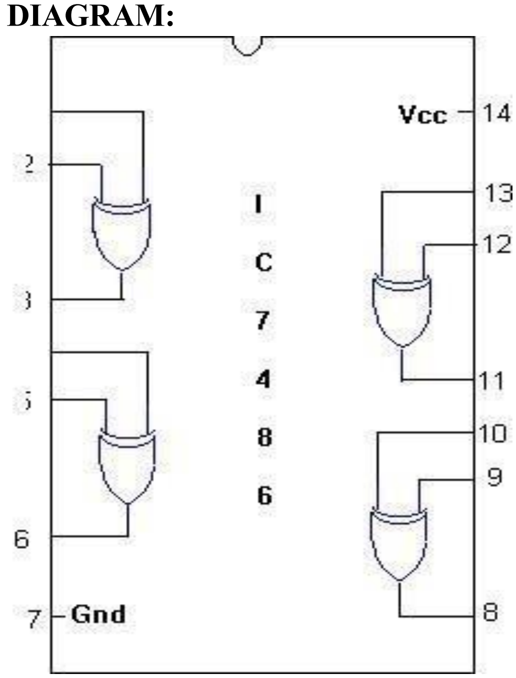
A	$\overline{A}$
0	1
1	0

XOR GATE:

SYMBOL:



PIN DIAGRAM:

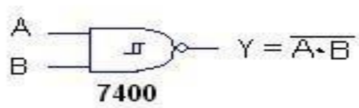


TRUTH TABLE :

A	B	$\overline{A}B + A\overline{B}$
0	0	0
0	1	1
1	0	1
1	1	0

2-INPUT NAND GATE:

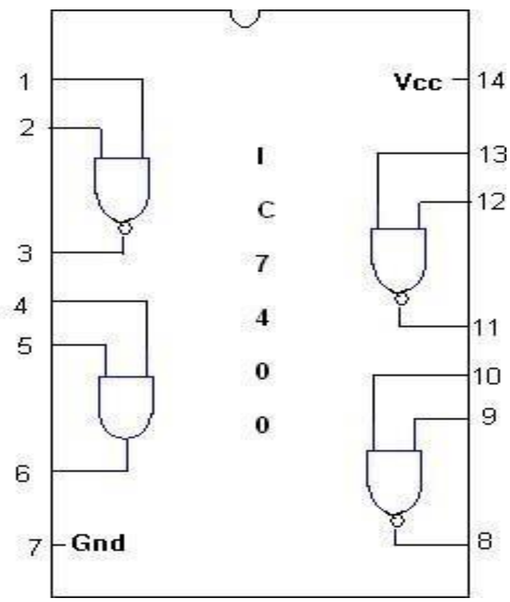
SYMBOL:



TRUTH TABLE

A	B	$\overline{A \cdot B}$
0	0	1
0	1	1
1	0	1
1	1	0

PIN DIAGRAM:



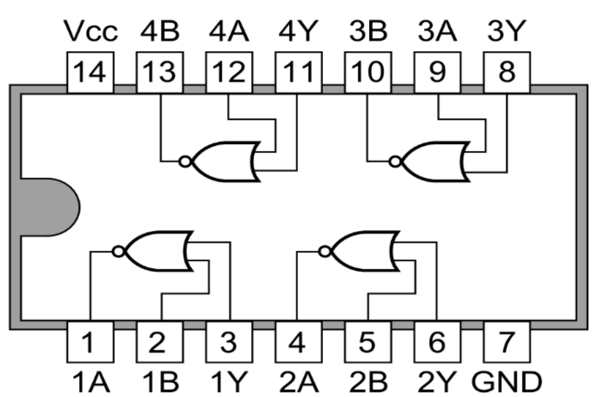
NOR GATE:

SYMBOL:

A	B	Q
0	0	1
0	1	0
1	0	0
1	1	0

PIN DIAGRAM:

7402 Quad 2-input NOR Gates



PROCEDURE:

Place the breadboard gently on the observation table.

I. Fix the IC which is under observation between the half shadow line of breadboard, so there is no shortage of voltage.
II. Connect the wire to the main voltage source (Vcc) whose other end is connected to last pin of the IC (14 place from the notch).
III. Connect the ground of IC (7th place from the notch) to the ground terminal provided on the digital lab kit.
IV. Give the input at any one of the gate of the ICs i.e. 1st, 2nd, 3rd, 4th gate by using connecting wires. (In accordance to IC provided).
V. Connect output pins to the led on digital lab kit. VII. Switch on the power supply.
VIII. If LED glows red then output is true, if it glows green output is false, which is numerically denoted as 1 and 0 respectively. The Color can change based on the IC manufacturer it's just verification of the Truth Table not the color change.

RESULT:

Thus the logic gates are studied and their truth tables were verified.

PRECAUTIONS:-

- All connections should be made neat and tight.
- Digital lab kits and ICs should be handled with utmost care.
- While making connections main voltage should be kept switched off.
- Never touch live and naked wires.

EXPERIMENT NO: -2

LEARNING OUTCOMES: -

By the end of Lab. Session, the students should be able to demonstrate knowledge of DeMorgan's laws and their significance in Boolean
algebra.

1. **Verification of Demorgan's theorem for 2 variables**

APPARATUS REQUIRED

SL NO.	COMPONENT	SPECIFICATION	QUANTITY
1	NOT GATE	IC 7404	1
2	OR GATE	IC 7432	1
3	AND GATE	IC 7408	1
2	BREAD BOARD	-	1
3	RESISTOR	220Ω	1
4	BATTERY	9V	1
5	LED	-	1
6	CONNECTING WIRE	-	AS PER REQUIREMENT

THEORY

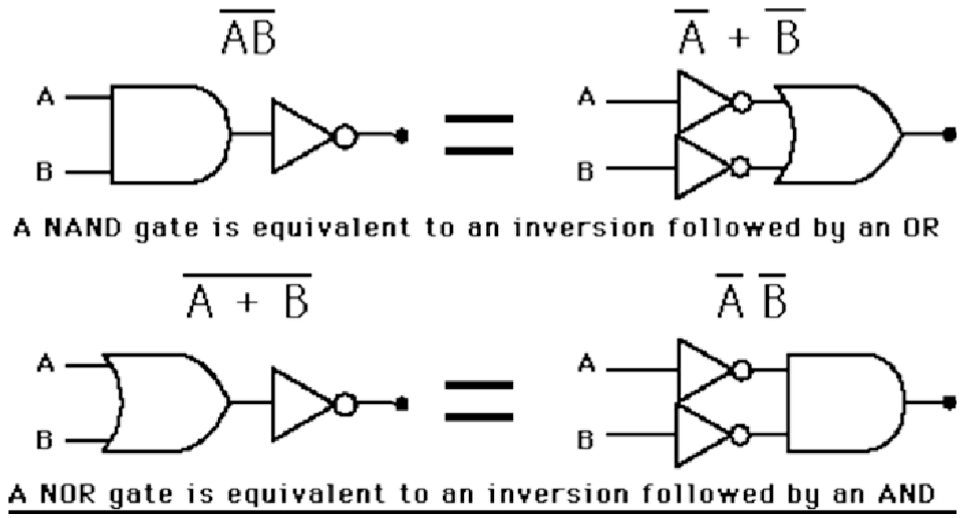
Theorem 1: The compliment of the product of two variables is equal to the sum of the compliment of each variable. Thus according to De-Morgan's laws or De-Morgan's theorem if A and B are the two variables or Boolean numbers. Then accordingly,

$$(\mathrm{A}\,.\,\mathrm{B})' = \bar{A} + B$$

Theorem 2:-

The compliment of the sum of two variables is equal to the product of the compliment of each variable. Thus according to De Morgan's theorem if A and B are the two variables then,

$$(\mathrm{A} + \mathrm{B})' = \bar{A}\,.\,B$$



De-Morgan's laws can also be implemented in Boolean algebra in the following steps:

1. While doing Boolean algebra at first replace the given operator. That is (+) is replaced with (.) and (.) is replaced with (+).
2. Compliment of each of the term is to be found.

PROCEDURE

1. Realize the Demorgan's theorem using logic gates.
2. Connect VCC and ground as shown in the pin diagram.
3. Make connections as per the logic gate diagram.
4. Apply the different combinations of input according to the truth tables and verify that the results are correct.

OBSERVATION [L=logic 0, H=logic 1]

Theorem 1

INPUT A	INPUT B	$\bar{\bar{\bar{A}}}.\bar{B}$	$\bar{A}+\bar{B}$
L	L		
L	H		
H	L		
H	H		

Theorem 2

INPUT A	INPUT B	$\bar{\bar{\bar{A}}}\bar{\bar{\bar{+}}}\bar{B}$	$\bar{A}.\bar{B}$
L	L		
L	H		
H	L		
H	H		

RESULT: The truth table of Demorgan's theorem for 2 variables is verified.

II. Implementation of the Given Boolean Function using Logic Gates.

AIM: To implement the given Boolean Function using Logic Gates.

$$Y = AB + A'B$$

APPARATUS REQUIRED

SL NO.	COMPONENT	SPECIFICATION	QUANTITY
1	NAND GATE	IC 7400	1
2	OR GATE	IC 7432	1
3	AND GATE	IC 7408	1
2	BREAD BOARD	-	1
3	RESISTOR	220Ω	1
4	BATTERY	9V	1
5	LED	-	1
6	CONNECTING WIRE	-	AS PER REQUIREMENT

THEORY

It is desirable to connect gates together in as few a number as possible to create desired output result given a fixed set of input conditions. Alternatively, it may be necessary to utilize one type of gate to produce several other logic functions. Purchasing one IC type in bulk quantity has the advantage of reducing the cost of these chips. Two theories are used to facilitate these objectives.

i) Boolean Algebra and

ii) Demorgan's Theorem

Boolean algebra utilizes rules based on logic gate operations.

Karnaugh's Map Reduction Technique

The implication of logic circuits is based on Karnaugh's map. There are two fundamental approaches in logic design.

i. Product of Sum method

ii. Sum of product method

The Sum of product solution results in NAND circuit, while the Product of Sum solution results in NOR circuit corresponding to a given truth table.

The designers can select either method, usually selects the simpler circuit because it costs less and more reliable.

The complexity of the digital logic gates can be implemented in a Boolean function. The map method provides a simple straightforward procedure for minimizing the Boolean functions. This map method is known as "Karnaugh's map".

The map is the diagram made up squares. Each square represents one minterm. Since any Boolean functions can be expressed as a sum of minterm. The simplest algebraic expression is any one from a sum of products or product of sums that has a minimum number of literals.

Two variable map has 4 minterms. It can be represented as

m0	m1

m2	m3

$\bar{x}\bar{y}$	$\bar{x}y$
$x\bar{y}$	xy

Two Variable Map

A three variable map has 8 minterms for the three variables. This map has 8 squares. Three variable map is represents as shown below.

m0	m1	m3	m2
m4	m5	m7	m6

$\bar{x}\bar{y}\bar{z}$	$\bar{x}\bar{y}\bar{z}$	$\bar{x}yz$	$\bar{x}y\bar{z}$
$x\bar{y}\bar{z}$	$x\bar{y}z$	xyz	$xy\bar{z}$

Three Variable Map

PROCEDURE

1. Position the IC's [7408, 7432, and 7404] properly on the solder less bread board.
2. Connect the pin 7 of all the chips to ground using patch chord provided.
3. Connect the pin 14 of all the chips to +5V supply using patch chord provided.
4. Input A signal and B signals are connected at pin no 1 and 2 of the IC 7408.
5. Input A signal is complemented by connecting pin 1 of the IC 7404.
6. The inverted input A and B signals are connected to the pin 4 and 5 of IC 7432.
7. The output (7408) pin 3 and 6 are concerted to input pin 1 and 2 of OR gate.

The output pin 3 of the OR gate give the Boolean function result.

OBSERVATION [L=logic 0, H=logic 1]]

INPUT A	**INPUT B**	$Y = AB + A'B$
L	**L**	
L	**H**	
H	**L**	
H	**H**	

RESULT: The given Boolean Function using Logic Gates is verified.

EXPERIMENT – 3

LEARNING OUTCOMES: -

By the end of Lab. Session, the students should be to derive the implementation of basic gates (NOT, AND, OR), NOR, XOR and XNOR using only NAND and NOR gates.

APPARATUS REQUIRED: -

logic trainer kit, NAND gates (IC 7400), NOR gates (IC 7402), wires.

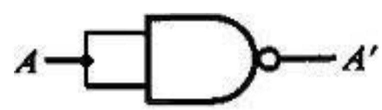
THEORY: -

NAND gate is actually a combination of two logic gates: AND gate followed by NOT gate. So its output is complement of the output of an AND gate.
This gate can have minimum two inputs; output is always one. By using only NAND gates, we can realize all logic functions: AND, OR, NOT, X-OR, X-NOR, NOR. So this gate is also called universal gate.

NAND gates as NOT gate:

A NOT produces complement of the input. It can have only one input, tie the inputs of a NAND gate together. Now it will work as a NOT gate. Its output is

$$Y = (A.A)'$$
$$Y = (A)'$$

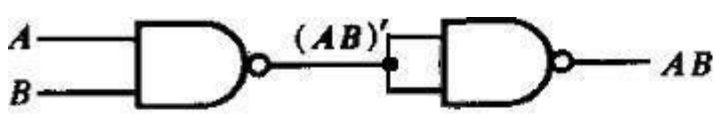


NOT (inverter)

NAND gates as AND gate:

A NAND produces complement of AND gate. So, if the output of a NAND gate is inverted, overall output will be that of an AND gate.

$$Y = ((A.B)')'$$
$$Y = (A.B)$$



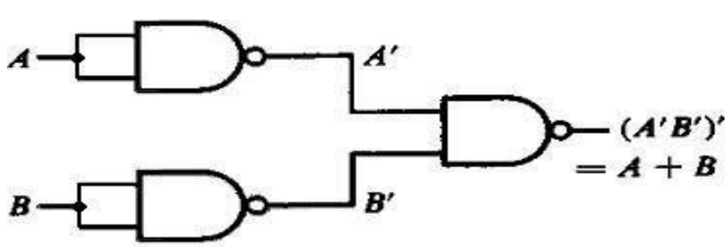
AND

NAND gates as OR gate:

From DeMorgan's theorems: $(A.B)' = A' + B'$

$$(A'.B')' = A'' + B'' = A + B$$

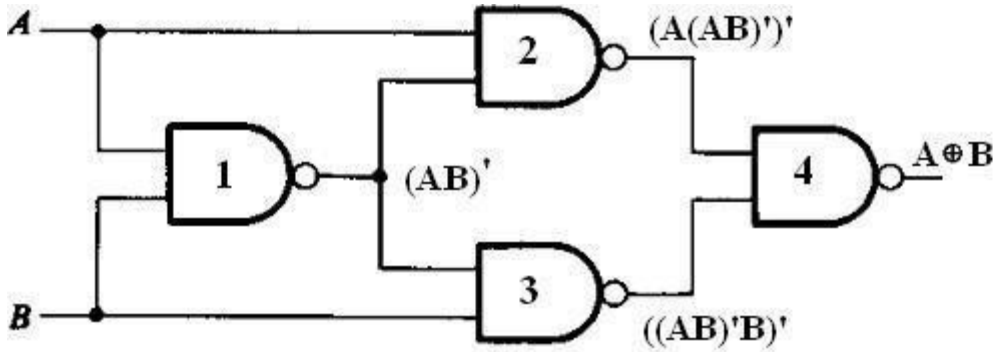
So, give the inverted inputs to a NAND gate, obtain OR operation at output.



OR

NAND gates as X-OR gate:

The output of a two input X-OR gate is shown by: Y = A'B + AB'. This can be achieved with the logic diagram shown in the below.



X-OR

Gate No.	Inputs	Output
1	A, B	(AB)'
2	A, (AB)'	(A (AB)')'
3	(AB)', B	(B (AB)')'
4	(A (AB)')', (B (AB)')'	A'B + AB'

Now the output from gate no. 4 is the overall output of the configuration.

Y = ((A (AB)')' (B (AB)')')'

= (A(AB)')'' + (B(AB)')''

= (A(AB)') + (B(AB)')

= (A(A' + B)') + (B(A' + B'))

= (AA' + AB') + (BA' + BB')

= (0 + AB' + BA' + 0)

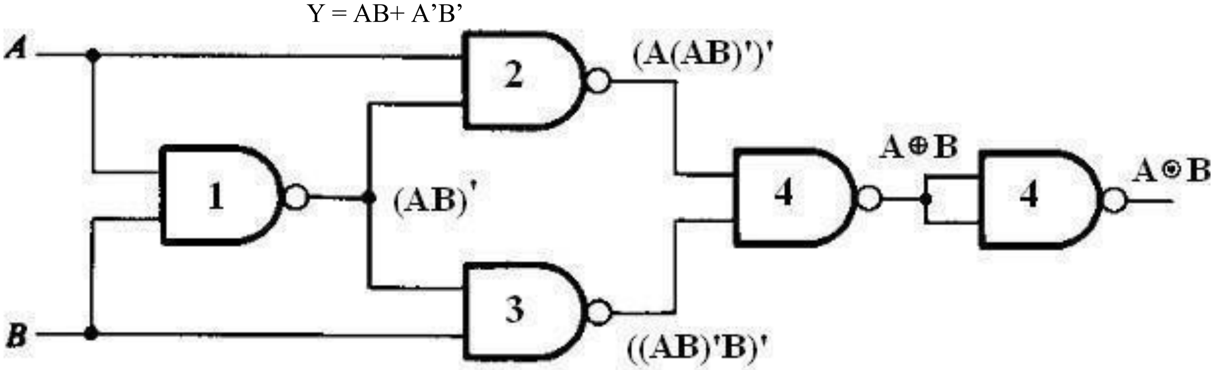
= AB' + BA' **Y**

= AB' + A'B

NAND gates as X-NOR gate:

X- NOR gate is actually X-OR gate followed by NOT gate. So give the output of X-OR gate to a NOT gate, overall output is that of an X-NOR gate.

Y = AB+ A'B'

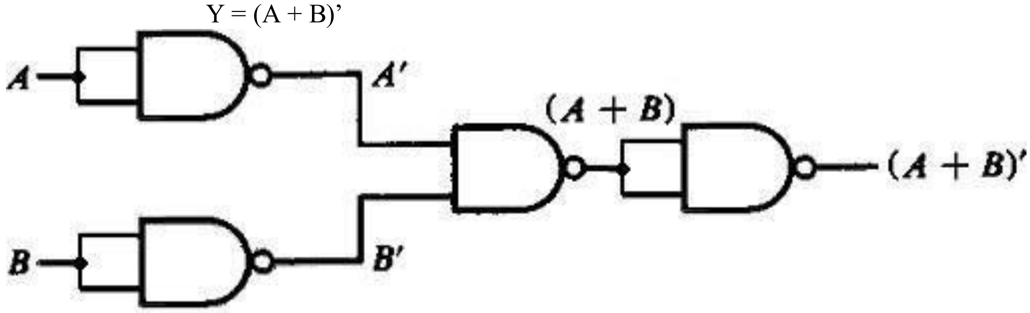


X-NOR

NAND gates as NOR gate

A NOR gate is an OR gate followed by NOT gate. So connect the output of OR gate to a NOT gate, overall output is that of a NOR gate.

Y = (A + B)’



NOR

PROCEDURE:

1. Connect the trainer kit to ac power supply.
2. Connect the NAND gates for any of the logic functions to be realized.
3. Connect the inputs of first stage to logic sources and output of the last gate to logic indicator.
4. Apply various input combinations and observe output for each one.
5. Verify the truth table for each input/ output combination.
6. Repeat the process for all logic functions. 7. Switch off the power supply.

THEORY:

NOR gate is actually a combination of two logic gates: OR gate followed by NOT gate. So its output is complement of the output of an OR gate.

This gate can have minimum two inputs; output is always one. By using only NOR gates, we can realize all logic functions: AND, OR, NOT, X-OR, X-NOR, NAND. So this gate is also called universal gate.

NOR gates as NOT gate:

A NOT produces complement of the input. It can have only one input, tie the inputs of a NOR gate together. Now it will work as a NOT gate. Its output is

Y = (A+A)’

Y = (A)’

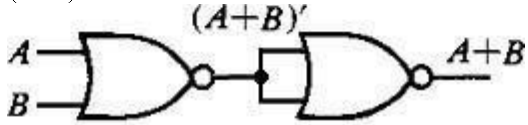
NOT

NOR gates as OR gate:

A NOR produces complement of OR gate. So, if the output of a NOR gate is inverted, overall output will be that of an OR gate.

Y = ((A+B)’)’

Y = (A+B)



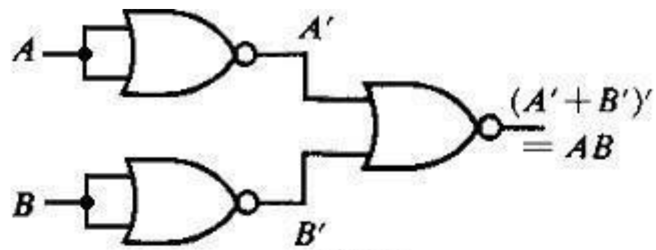
OR

NOR gates as AND gate:

From DeMorgan's theorems: (A+B)' = A'B'

(A'+B')' = A''B'' = AB

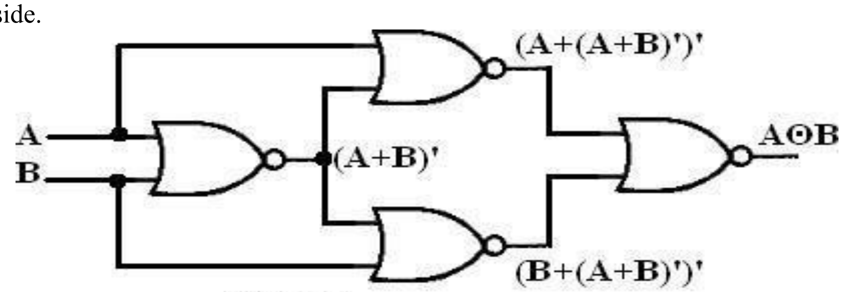
So, give the inverted inputs to a NOR gate, obtain AND operation at output.



AND

NOR gates as X-NOR gate:

The output of a two input X-NOR gate is shown by: Y = AB + A'B'. This can be achieved with the logic diagram shown in the left side.



X-NOR

Gate No.	Inputs	Output
1	A, B	(A + B)'
2	A, (A + B)'	(A + (A+B)')'
3	(A + B)', B	(B + (A+B)')'
4	(A + (A + B)')', (B + (A+B)')'	AB + A'B'

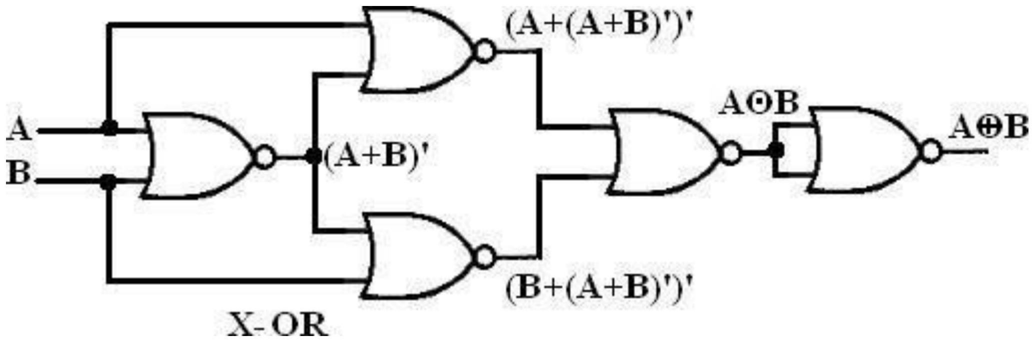
Now the output from gate no. 4is the overall output of the configuration.

$$
\begin{aligned}
Y &= ((A + (A+B)')' (B +(A+B)')')' \\
&= (A+(A+B)')''.(B+(A+B)')'' \\
&= (A+(A+B)').(B+(A+B)') \\
&= (A+A'B').(B+A'B') \\
&= (A + A').(A + B').(B+A')(B+B') \\
&= 1.(A+B').(B+A').1 \\
&= (A+B').(B+A') \\
&= A.(B + A') +B'.(B+A') \\
&= AB + AA' +B'B+B'A' \\
&= AB + 0 + 0 + B'A' \\
&= AB + B'A' \\
\mathbf{Y} &= AB + A'B'
\end{aligned}
$$

NOR gates as X-OR gate:

X- OR gate is actually X-NOR gate followed by NOT gate. So give the output of X-NOR gate to a NOT gate, overall output is that of an X-OR gate.

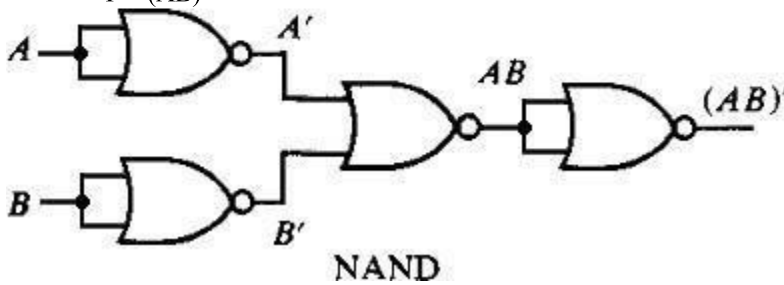
Y = A'B+ AB'



NOR gates as NAND gate:

A NAND gate is an AND gate followed by NOT gate. So connect the output of AND gate to a NOT gate, overall output is that of a NAND gate.

Y = (AB)'



PROCEDURE:

1. Connect the trainer kit to ac power supply.
2. Connect the NOR gates for any of the logic functions to be realised.
3. Connect the inputs of first stage to logic sources and output of the last gate to logic indicator.
4. Apply various input combinations and observe output for each one.
5. Verify the truth table for each input/ output combination.
6. Repeat the process for all logic functions. 7. Switch off the ac power supply.

RESULT:

NAND & NOR are verified as universal gates successfully.

PRECAUTIONS:-

- All connections should be made neat and tight.
- Digital lab kits and ICs should be handled with utmost care.
- While making connections main voltage should be kept switched off.
- Never touch live and naked wires.

EXPERIMENT – 4

LEARNING OUTCOMES: -

By the end of Lab. Session, the students should be able to design Half Adder and Full Adder circuits using basic logic gates (AND, OR, XOR) and differentiate between Half Adder and Full Adder in terms of inputs, outputs, and functionality.

APPARATUS REQUIRED: -

logic trainer kit, NAND gates (IC 7400), XOR gates (IC 7486), AND gates (IC 7408), wires.

THEORY:

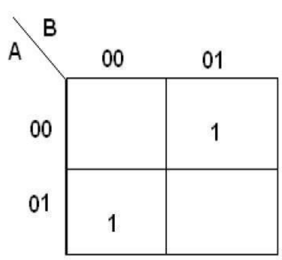
Half adder:

A half adder has two inputs for the two bits to be added and two outputs one from the sum ' S' and other from the carry 'C' into the higher adder position. Above circuit is called as a carry signal from the addition of the less significant bits sum from the X-OR Gate the carry out from the AND gate.

Truth Table for Half Adder

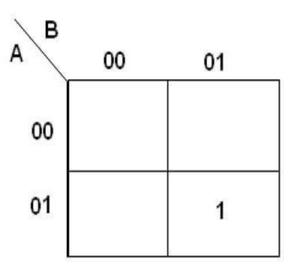
INPUT		OUTPUT	
A	**B**	**SUM**	**CARRY**
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

K-Map for SUM:



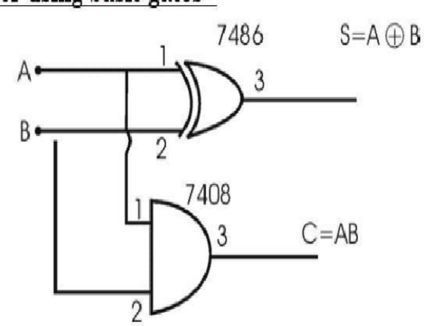
SUM = A'B + AB'

K-Map for CARRY:



CARRY = AB

Half Adder using basic gates:-

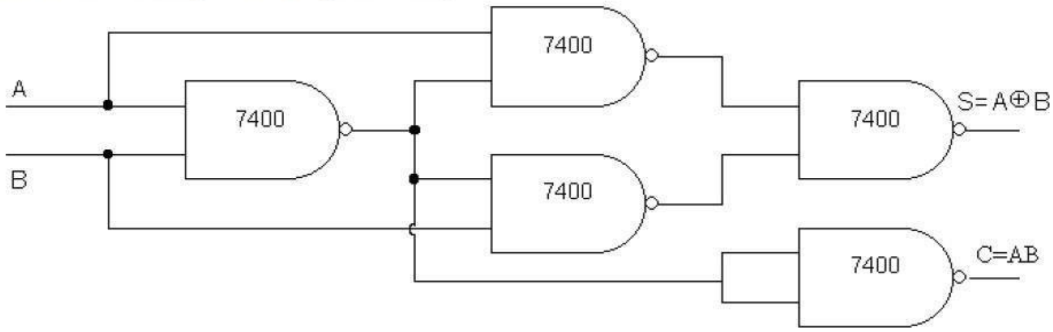


$$S = \bar{A}B + A\bar{B}$$

$$S = A \oplus B$$

$$C = AB$$

Half Adder using NAND gates only:-



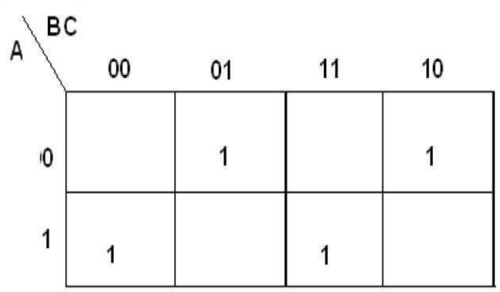
Full Adder:

A full adder is a combinational circuit that forms the arithmetic sum of input; it consists of three inputs and two outputs. A full adder is useful to add three bits at a time but a half adder cannot do so. In full adder sum output will be taken from X-OR Gate, carry output will be taken from OR Gate.

Truth Table for Full Adder

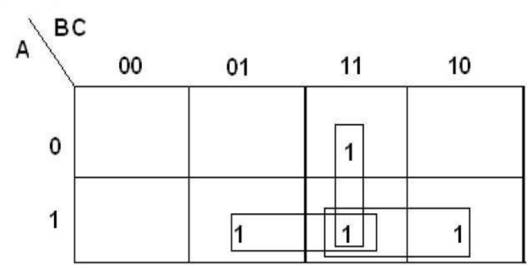
INPUT			OU PUT	
A	**B**	**C**	**SUM**	**CARRY**
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

K-Map for SUM:



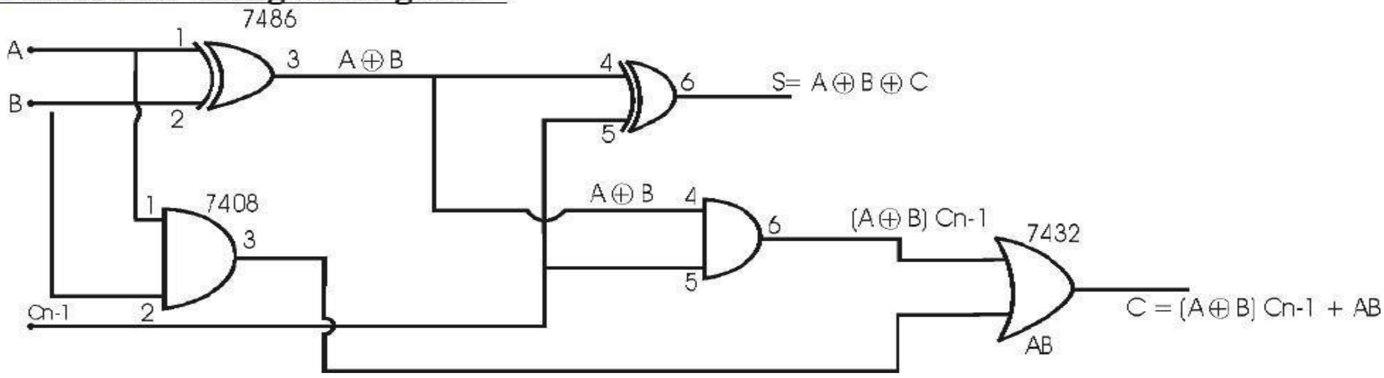
SUM = A'B'C + A'BC' + ABC' + ABC

K-Map for CARRY:

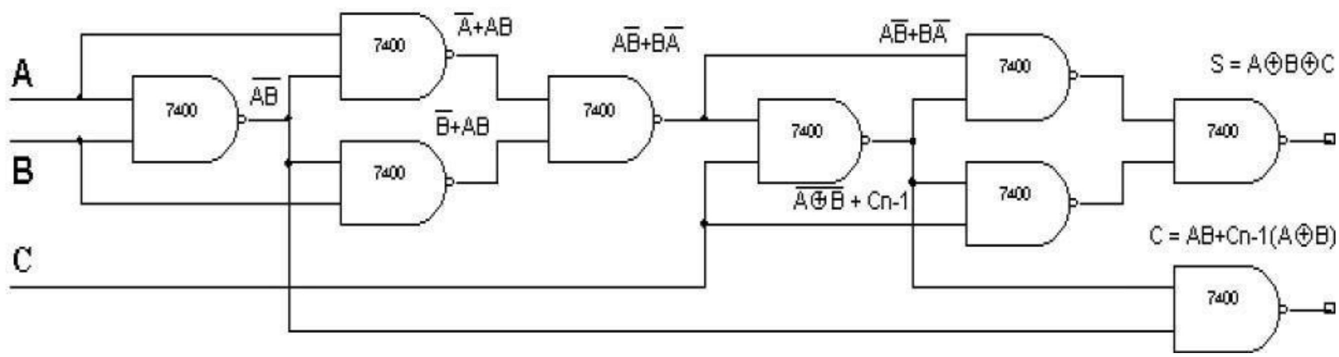


CARRY = AB + BC + AC

Full Adder using basic gates:-



Full Adder using NAND gates only:-



PROCEDURE:

(i) Connections are given as per circuit diagram.
(ii) Logical inputs are given as per circuit diagram. (iii) Observe the output and verify the truth table.

RESULT:

Thus the half adder &full adder was designed and their truth table is verified.

PRECAUTIONS:

- All connections should be made neat and tight.
- Digital lab kits and ICs should be handled with utmost care.
- While making connections main voltage should be kept switched off.
- Never touch live and naked wires.

EXPERIMENT – 5

LEARNING OUTCOMES: -

By the end of Lab. Session, the students will be able to derive Boolean expressions for *Difference* and *Borrow* outputs of Half and Full Subtractors and Differentiate between Half and Full Subtractor in terms of inputs, outputs, and borrow concept.

APPARATUS REQUIRED: -

logic trainer kit, NAND gates (IC 7400), XOR gates (IC 7486), AND gates (IC 7408), NOT gates (IC 7404), connecting wires.

THEORY:

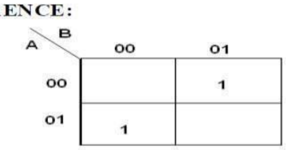
Half Subtractor:

The half subtractor is constructed using X-OR and AND Gate. The half subtractor has two input and two outputs. The outputs are difference and borrow. The difference can be applied using X-OR Gate, borrow output can be implemented using an AND Gate and an inverter.

Truth Table for Half Subtractor

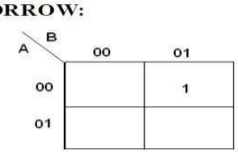
INPUT		OUTPUT	
A	**B**	**Bo**	**Diff**
0	0	0	0
0	1	1	1
1	0	0	1
1	1	0	0

K-Map for DIFFERENCE:



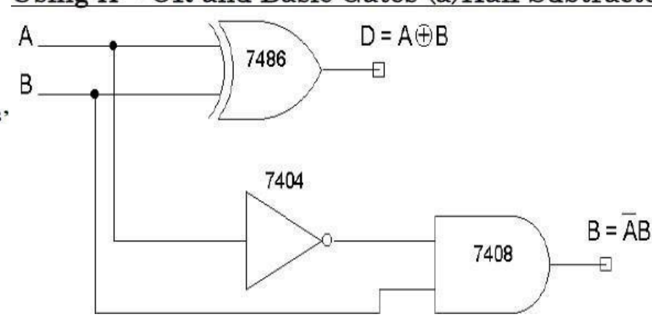
DIFFERENCE = A'B + AB'

K-Map for BORROW:

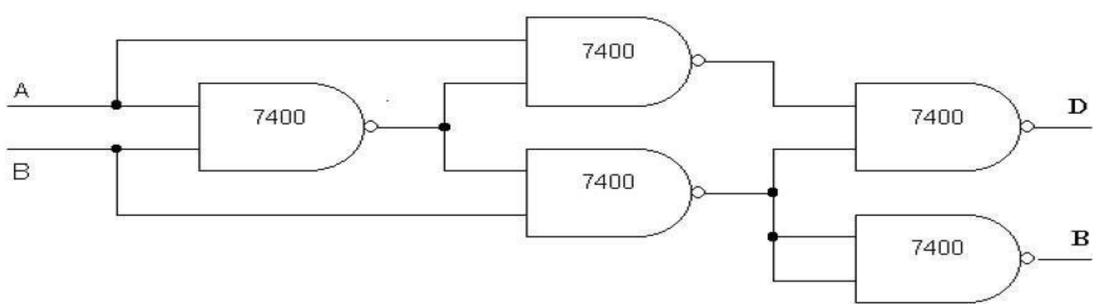


BORROW = A'B

Using X – OR and Basic Gates (a)Half Subtractor



(ii) Using only NAND gates (a) Half subtractor



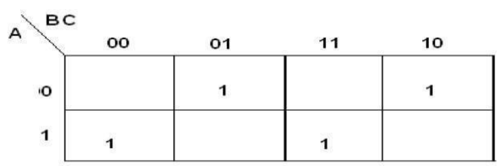
Full Subtractor:

The full subtractor is a combination of X-OR, AND, OR, NOT Gates. In a full subtractor the logic circuit should have three inputs and two outputs. The two half subtractor put together gives a full subtractor .The first half subtractor will be C and A B. The output will be difference output of full subtractor. The expression AB assembles the borrow output of the half subtractor and the second term is the inverted difference output of first X-OR.

Truth Table for Full Subtractor

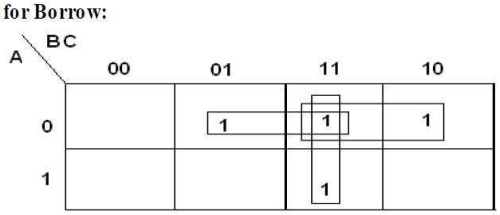
INPUT			OUTPUT	
A	**B**	**C**	**Bo**	**Diff**
0	0	0	0	0
0	0	1	1	1
0	1	0	1	1
0	1	1	1	0
1	0	0	0	1
1	0	1	0	0
1	1	0	0	0
1	1	1	1	1

K-Map for Difference:

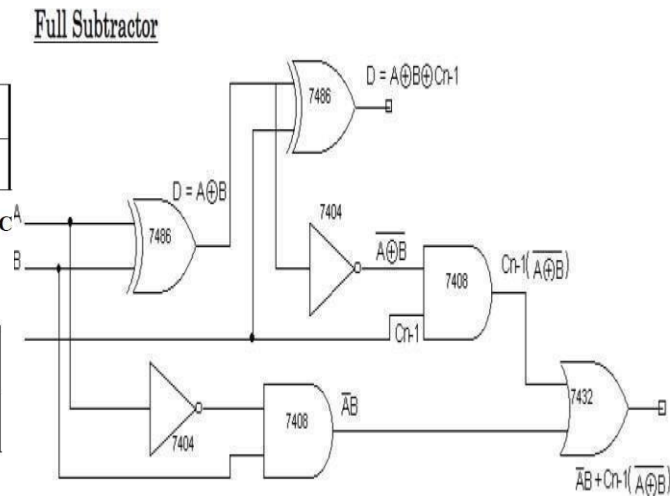


Difference = A'B'C + A'BC' + AB'C' + ABC

K-Map for Borrow:



Borrow = A'B + BC + A'C



PROCEDURE:

(i) Connections are given as per circuit diagram.

(ii) Logical inputs are given as per circuit diagram. (iii) Observe the output and verify the truth table.

RESULT:

Thus the half subtractor and full subtractor was designed and their truth table is verified.

PRECAUTIONS:

- All connections should be made neat and tight.
- Digital lab kits and ICs should be handled with utmost care.
- While making connections main voltage should be kept switched off.
- Never touch live and naked wires.

EXPERIMENT – 6

LEARNING OUTCOMES: -

By the end of Lab. Session, the students should be able to identify pin configuration and function of the IC-74151 (Multiplexer) and realize how a multiplexer can implement combinational logic circuits

APPARATUS REQUIRED: -

- logic trainer kit, IC-74151
- Connecting wires.
- Breadboard.
- Multiple power supply

THEORY:

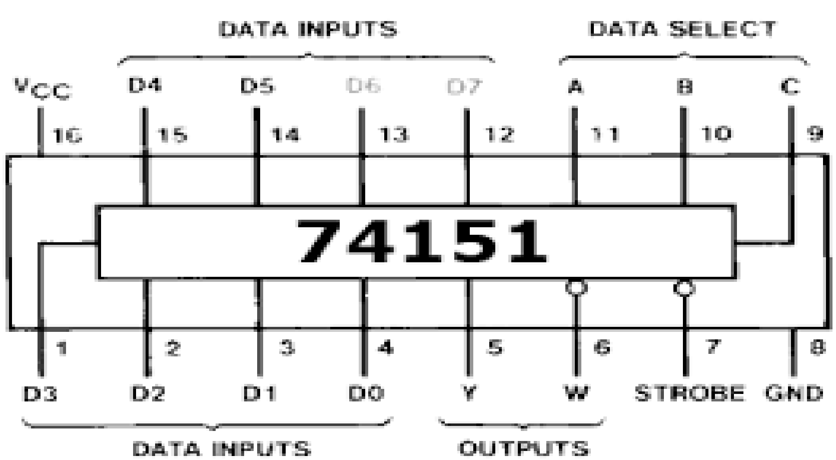
The TTL/MSI SN54/74LS151 is a high speed 8-input Digital Multiplexer. It provides, in one package, the ability to select one bit of data from up to eight sources. The LS151 can be used as a universal function generator to generate any logic function of four variables.

- Multiplexer is a combinational circuit that is one of the most widely used in digital design.
- The multiplexer is a data selector which gates one out of several inputs to a single o/p. It has n data inputs & one o/p line & m select lines where 2^m= n shown in fig a.
- Depending upon the digital code applied at the select inputs one out of n data input is selected & transmitted to a single o/p channel.
- Normally strobe (G) input is incorporated which is generally active low which enables the multiplexer when it is LOW. Strobe i/p helps in cascading.
- IC 74151A is an 8: 1 multiplexer which provides two complementary outputs Y & Y. The o/p Y is same as the selected i/p & Y is its complement. The n: 1 multiplexer can be used to realize a m variable function. (2^m= n, m is no. of select inputs)

8:1 Multiplexer:

It has eight data inputs D0 to D7, three select inputs S0 to S2, an enable input and one output.

IC PINOUT:

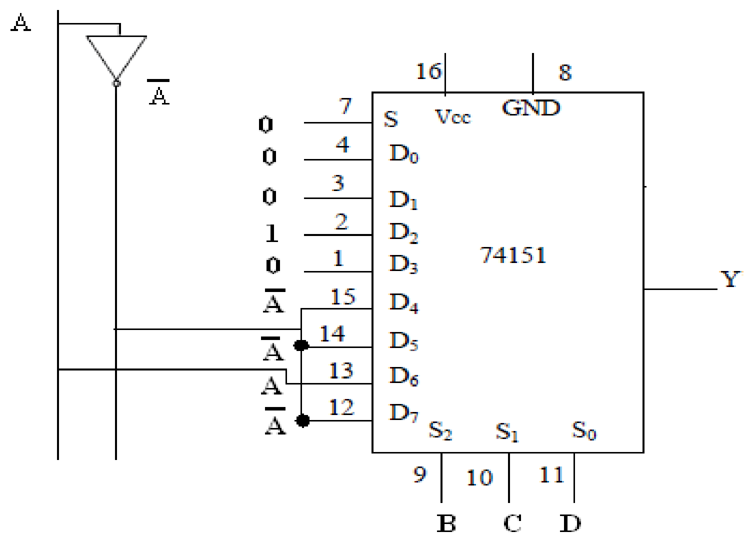


Block diagram &Truth Table:

IC 74151 (8-input Digital Multiplexer)

Block diagram of n:1 MUX **Truth Table**

Logic Diagram



Data input and select input lines

It has eight data inputs D0 to D7, three select inputs S0 to S2, an enable input and one output.

8:1 MUX

Sr.	D0	D1	D2	D3	D4	D5	D6	D7	S2	S1	S0	$\overline{E}$	$\overline{Y}$	Y
0	1	0	0	0	0	0	0	0	0	0	0	0	0	D0
1	0	1	0	0	0	0	0	0	0	0	1	0	0	D1
2	0	0	1	0	0	0	0	0	0	1	0	0	0	D2
3	0	0	0	1	0	0	0	0	0	1	1	0	0	D3
4	0	0	0	0	1	0	0	0	1	0	0	0	0	D4
5	0	0	0	0	0	1	0	0	1	0	1	0	0	D5
6	0	0	0	0	0	0	1	0	1	1	0	0	0	D6
7	0	0	0	0	0	0	0	1	1	1	1	0	0	D7

PROCEDURE: -

1) Assemble the circuit on bread board, as per above diagram.
2) Give the logical inputs and check for the proper output, as per the truth table.

CONCLUSION:

Thus, we have implemented four input Boolean function using 8:1 multiplexer of IC 74151.

PRECAUTIONS:

- All connections should be made neat and tight.
- Digital lab kits and ICs should be handled with utmost care.
- While making connections main voltage should be kept switched off.
- Never touch live and naked wires.

EXPERIMENT – 7

LEARNING OUTCOMES: -

By the end of Lab. Session, the students should be able to explain the working principle of a De-Multiplexer (De-MUX) as a data distributor and understand the role of select lines, enable pins, input, and outputs in IC-74138 (3-to-8 line De-MUX/decoder).

APPARATUS REQUIRED: -

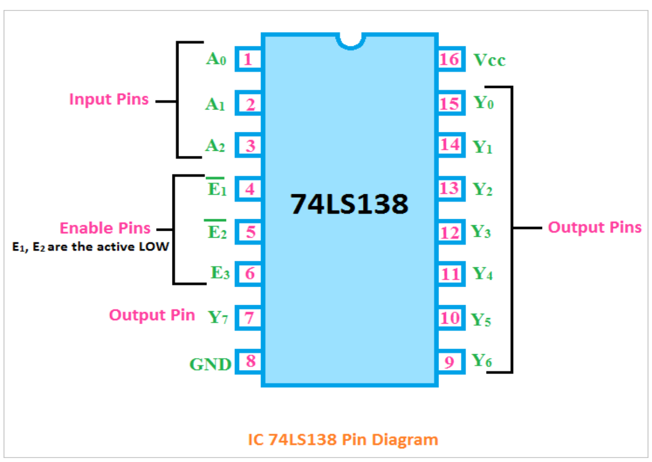
Logic trainer kit, IC- 74138, wires.

THEORY:

A Demultiplexer performs the reverse operation of a Multiplexer. It accepts a single input and distributes it over several outputs. The SELECT input code determines to which output the data input will be transmitted. The Demultiplexer becomes enabled when the strobe signal is active LOW.
This circuit can also be used as binary-to-decimal decoder with binary inputs applied at the select input lines and the output will be obtained on the corresponding line. These devices are available as 2-line-to-4-line decoder, 3-line-to- 8-line decoder, 4-line-to-16-line decoder. The output of these devices is active LOW. Also there is an active low enable/data input terminal available. Figure below shows the block diagram of a Demultiplexer.
We have three input pins which are actively in high state and are classified as I_2, I_1 and I_0. The outputs are actively in low state and are eight in number and are classified as O_7*, O_6*, …, O_0*. A power supply of +5 V DC is needed by the chip and is Grounded.

PIN CONFIGURATION:

As you see in the below pin diagram, the IC 74LS138 has a total of sixteen pins.



IC 74138 Pin Diagram

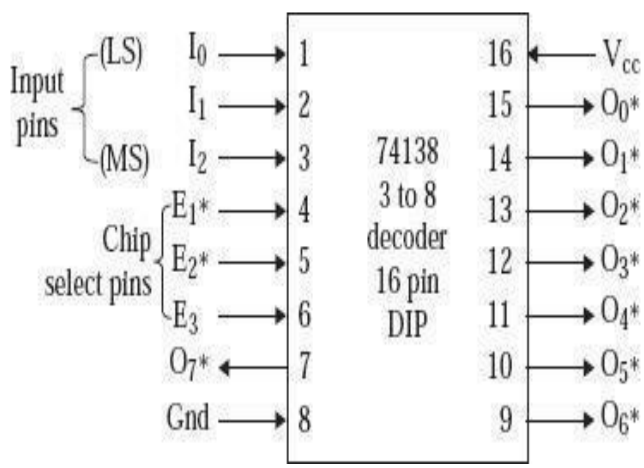
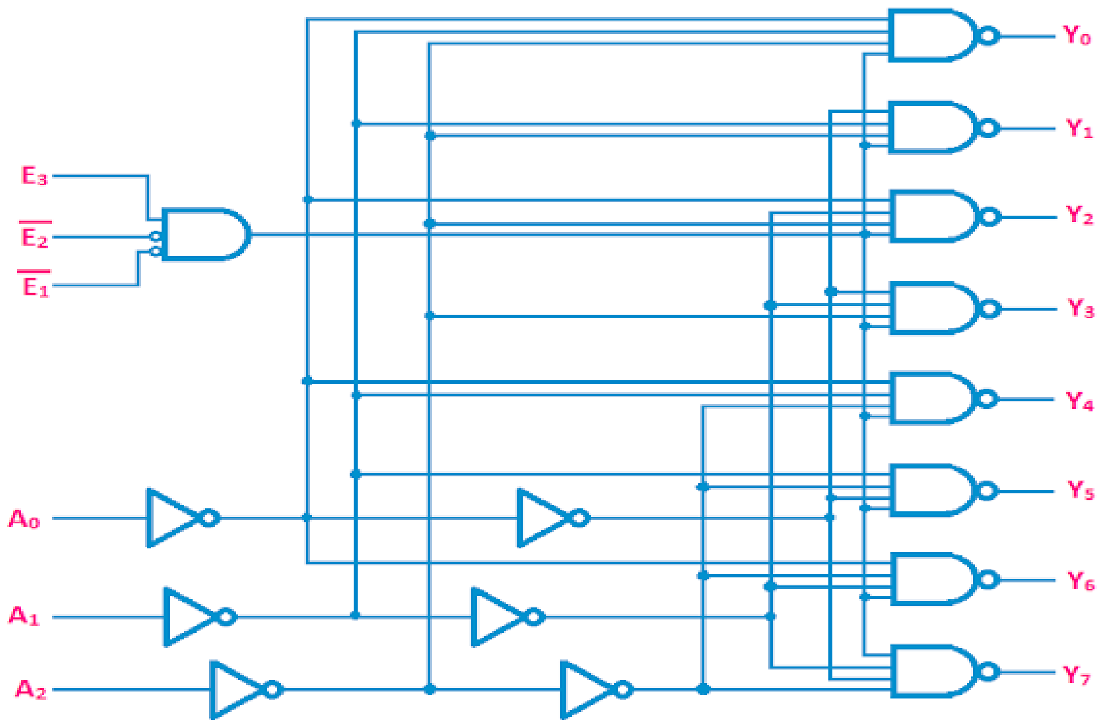


Diagram in functional mode of 74138

1. The pin no. 8 and 16 are the ground and Vcc respectively for the power input.
2. There are a total of three input pins (pin no. 1, 2, 3). They are denoted by A0, A1, A2. So, the IC 74LS138 can take three binary input signals.
3. There are three enable input pins E1, E2, E3 (pin no. 4, 5, 6). E1 and E2 are the active LOW pins that mean when low signals are applied to those pins they will be active
4. Pin no. 7 and 9 to 15 are the output pins.

Logical Diagram of IC 74138

The logical circuit of the IC 74138 is made using NOT Gate, and AND Gates. You can see the circuit diagram in the below figure.



Logical Diagram of IC 74138

IC 74138 Truth Table

Here the truth table of IC 74138 is given below.

INPUTS						OUTPUTS							
$\overline{E_1}$	$\overline{E_2}$	E_3	A_0	A_1	A_2	Y_0	Y_1	Y_2	Y_3	Y_4	Y_5	Y_6	Y_7
H	X	X	X	X	X	H	H	H	H	H	H	H	H
X	H	X	X	X	X	H	H	H	H	H	H	H	H
X	X	L	X	X	X	H	H	H	H	H	H	H	H
L	L	H	L	L	L	L	H	H	H	H	H	H	H
L	L	H	H	L	L	H	L	H	H	H	H	H	H
L	L	H	L	H	L	H	H	L	H	H	H	H	H
L	L	H	H	H	L	H	H	H	L	H	H	H	H
L	L	H	L	L	H	H	H	H	H	L	H	H	H
L	L	H	H	L	H	H	H	H	H	H	L	H	H
L	L	H	L	H	H	H	H	H	H	H	H	L	H
L	L	H	H	H	H	H	H	H	H	H	H	H	L

H - High, L - Low, X - Don't Care

IC 74138 Truth Table

PROCEDURE: -

1) Assemble the circuit on bread board, as per above Pin diagram.
2) Give the logical inputs and check for the proper output, as per the truth table.

CONCLUSION:

Hence, verified the Demultiplexer operation using IC-74138.

PRECAUTIONS:

- All connections should be made neat and tight.
- Digital lab kits and ICs should be handled with utmost care.
- While making connections main voltage should be kept switched off.
- Never touch live and naked wires.

EXPERIMENT – 8

LEARNING OUTCOMES: -

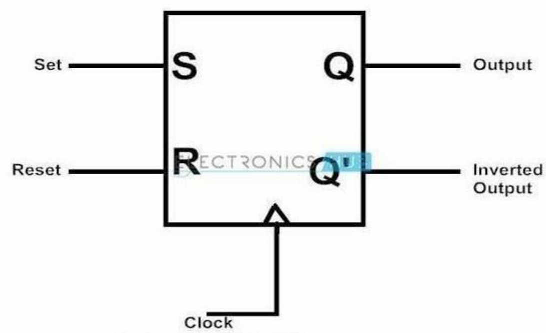
By the end of Lab. Session, the students should be able to realize the construction of the equivalent design of S-R flip-flop using NOR/NAND gates as memory elements in sequential circuits

APPARATUS REQUIRED: -

Logic trainer kit, NAND gates ICs- 7400, NOR gates ICs-7402, wires.

THEORY:

The SR flip-flop is one of the fundamental parts of the sequential circuit logic. SR flip-flop is a memory device and a binary data of 1 – bit can be stored in it. SR flip-flop has two stable states in which it can store data in the form of either binary zero or binary one. Like all flip-flops, an SR flip-flop is also an edge sensitive device.
SR flip–flop is one of the most vital components in digital logic and it is also the most basic sequential circuit that is possible. The S and R in SR flip-flop means 'SET' and 'RESET' respectively. Hence it is also called Set–Reset flip- flop. The symbolic representation of the SR Flip Flop is shown below.



Working Principle:

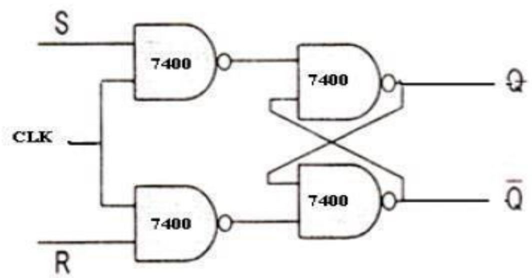
SR flip-flop works during the transition of clock pulse either from low to high or from high to low (depending on the design) i.e. it can be either positive edge triggered or negative edge triggered.
For a positive edge triggered SR flip-flop, suppose, if S input is at high level (logic 1) and R input is at low level (logic 0) during a low to high transition on clock pulse, then the SR flip-flop is said to be in SET state and the output of the SR flip-flop is SET to 1.
For the same clock situation, if the R input is at high level (logic 1) and S input is at low level (logic 0), then the SR flip-flop is said to be in RESET state and the output of the SR flip-flop is RESET to 0.

The SR flip-flops can be designed by using logic gates like NOR gates and NAND gates.

S-R Flip-Flop Using NAND Gate

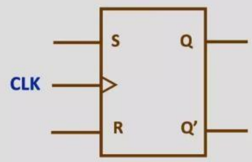
SR flip flop can be designed by cross coupling of two NAND gates. It is an active low input SR flip-flop. The circuit of SR flip-flop using NAND gates is shown in below figure:

R-S flip-flop using NAND gates



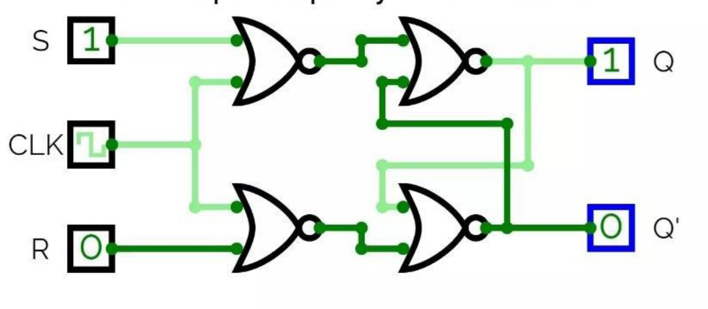
SR Flip-Flop Truth Table

Symbol



S	R	Q_{n+1}
0	0	Q_n
0	1	0
1	0	1
1	1	Indeterminate

SR Flip Flop By NOR Gate



PROCEDURE:

I. Connections are given as per circuit diagram.
II. Logical inputs are given as per circuit diagram.
III. Observe the output and verify the truth table.

RESULT:

Design of S-R Flip flop using NAND & NOR gates was verified successfully.

PRECAUTIONS:

- All connections should be made neat and tight.
- Digital lab kits and ICs should be handled with utmost care.
- While making connections main voltage should be kept switched off.
- Never touch live and naked wires.

EXPERIMENT – 9

LEARNING OUTCOMES: -

By the end of Lab. Session, apply various input combinations to a J-K flip-flop (7476) and observe corresponding output (Q, Q') experimentally.

APPARATUS REQUIRED: -

1. Logic trainer kit,
2. Flip-flop
3. ICs- 7476, wires.

THEORY:

The JK flip-flop is the modified version of SR flip-flop with no invalid state; i.e. the state J=K=1 is not forbidden. It works such that J serves as set input and K serves as reset. The only difference is that for the combination J=K=1 this flip-flop; now performs an action: it inverts its state.

JK Flip-Flop: Circuit, Truth Table

Truth Table			
Q	**J**	**K**	**Q(t+1)**
0	**0**	**0**	**0**
0	**0**	**1**	**0**
0	**1**	**0**	**1**
0	**1**	**1**	**1**
1	**0**	**0**	**1**
1	**0**	**1**	**0**
1	**1**	**0**	**1**
1	**1**	**1**	**0**

The flip-flop is constructed in such a way that the output Q is ANDed with K and CP. This arrangement is made so that the flip-flop is cleared during a clock pulse only if Q was previously 1. Similarly, Q' is ANDed with J and CP, so that the flip-flop is cleared during a clock pulse only if Q' was previously 1.

When J = K = 0

When both J and K are 0, the clock pulse has no effect on the output and the output of the flip-flop is the same as its previous value. This is because when both the J and K are 0, the output of their respective AND gate becomes 0.

When J=0, K=1

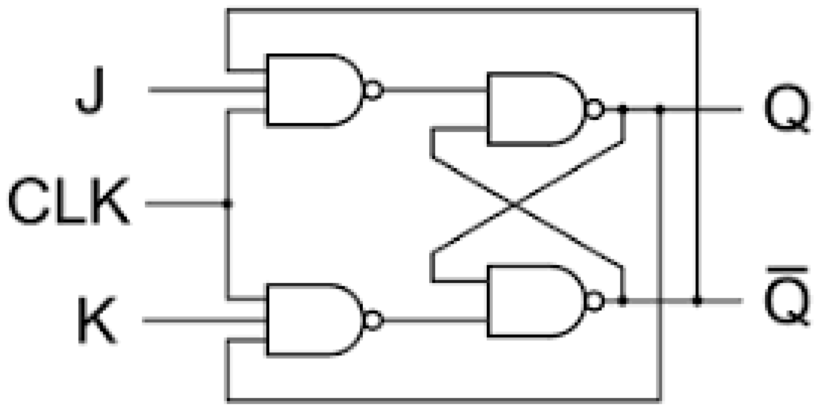
When J=0, the output of the AND gate corresponding to J becomes 0 (i.e.) S=0 and R=1. Therefore, Q' becomes 0. This condition will reset the flip-flop. This represents the RESET state of Flip-flop.

When J=1, K=0

In this case, the AND gate corresponding to K becomes 0(i.e.) S=1 and R=0. Therefore, Q becomes 0. This condition will set the Flip-flop. This represents the SET state of Flip-flop.

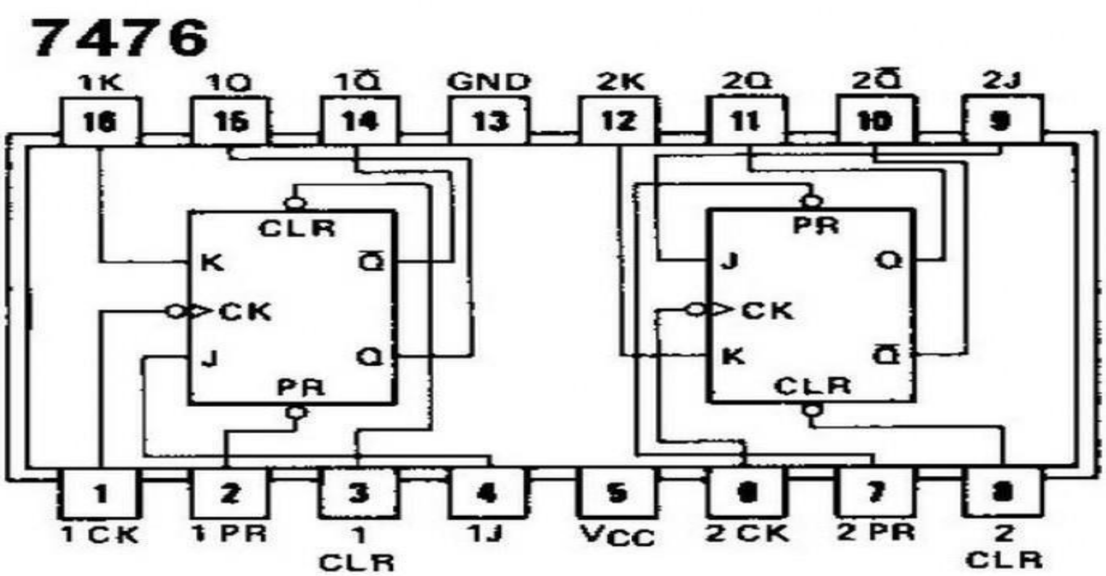
When J=K=1

Consider the condition of CP=1 and J=K=1. This will cause the output to complement again and again. This complement operation continues until the Clock pulse goes back to 0. Since this condition is undesirable, we have to find a way to eliminate this condition. This undesirable behavior can be eliminated by Edge triggering of JK flip- flop or by using master slave JK Flip-flops.



JK Flip-Flop: Circuit

JK Flip-Flop: Pin Out



PROCEDURE:

I. Connections are given as per circuit diagram.
II. Logical inputs are given as per circuit diagram.
III. Observe the output and verify the truth table.

RESULT:

Thus the J-K Flip flop was designed and truth table is verified.

PRECAUTIONS:

- All connections should be made neat and tight.
- Digital lab kits and ICs should be handled with utmost care.
- While making connections main voltage should be kept switched off.
- Never touch live and naked wires.

EXPERIMENT – 10

LEARNING OUTCOMES: -

By the end of Lab. Session, the students should be able to use various combinations to verify the truth table of a D and T flip-flop (JK-7476, NOT-7404, D-7474).

(a) Verify the truth table of a D flip-flop.

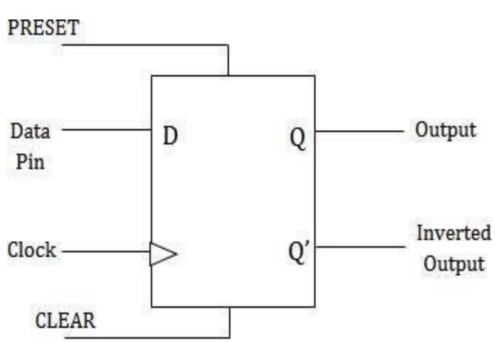
APPARATUS REQUIRED: -

Logic trainer kit, Flip-flop ICs- 7474, wires.

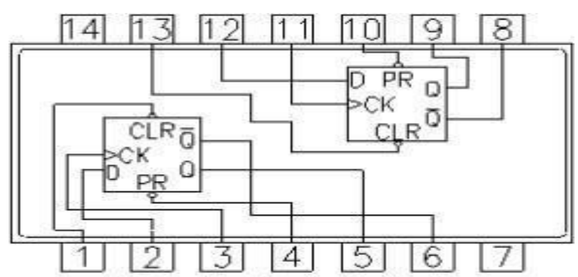
THEORY:

D flip flop also called as delay flip flop where it can be used to introduce a delay in the digital circuit by changing the propagation delay of the flip flop. Here the input data bit at D will reflects at the output after a certain propagation delay.

D Flip-Flop: Circuit, Truth Table, Pin Out



Symbol: D Flip-flop



7474
Dual D Flip-Flop
with Preset and Clear

Truth Table of D Flip-Flop				Characteristic Table of D Flip-Flop		
Clock	**INPUT**	**OUTPUT**		**D**	**Q**	**Q'**
	D	**Q**	0	0	0	**0**
LOW	X	0	0	1	0	0
HIGH	0	0	1	0	1	1
HIGH	1	1	1	1	1	1

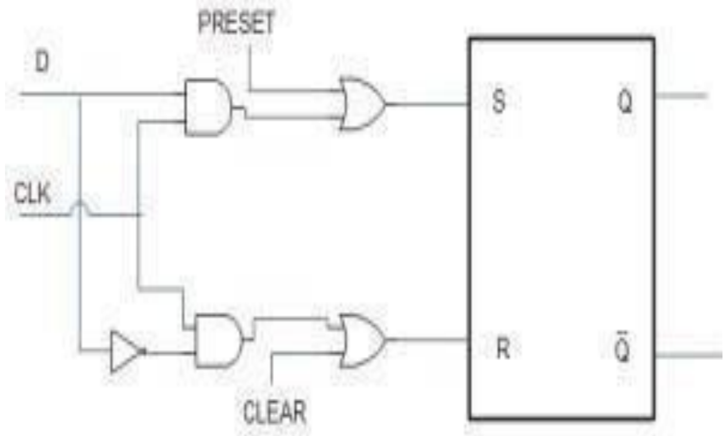
Characteristic Equation:

$$Q' = D\,Q' + D\,Q$$
$$Q' = D$$

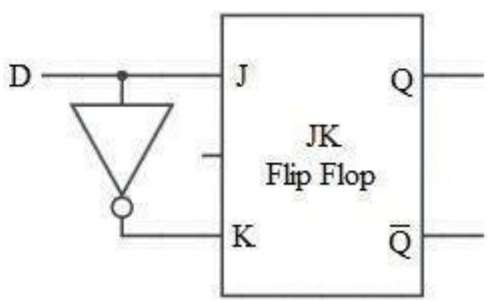
D Flip Flop with PRESET and CLEAR

PRESET is the input to the D flip flop which sets the output data to High i.e. 1. and CLEAR is also an input which clears the output data or output state. A high PRESET forces Q to 1; a high CLEAR resets Q to 0. Figure shows clocked flip flop with PRESET and CLEAR inputs.

Clocked D Flip-Flop with PRESET and CLEAR



Alternate Circuit



In the above circuit irrespective of the AND gates out, if the PRESET input is high the OR gate out directly sets the S input which makes Q to 1and in the same way if CLEAR input is high it resets the Q to 1.

PROCEDURE:

I. Connections are given as per circuit diagram.
II. Logical inputs are given as per circuit diagram.
III. Observe the output and verify the truth table.

RESULT:

Thus the D Flip flop was designed and their truth table is verified.

PRECATIONS:

All connections should be made neat and tight.

- Digital lab kits and ICs should be handled with utmost care.
- While making connections main voltage should be kept switched off.
- Never touch live and naked wires.

(b) Verify the truth table of a T flip-flop (JK 7476, NOT)

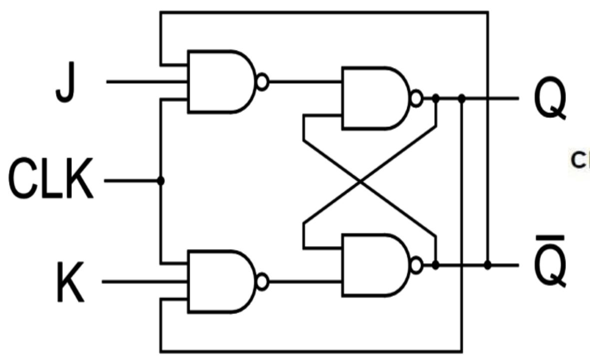
APPARATUS REQUIRED: -

Logic trainer kit, Flip-flop ICs- 7476, NOT 7404, wires.

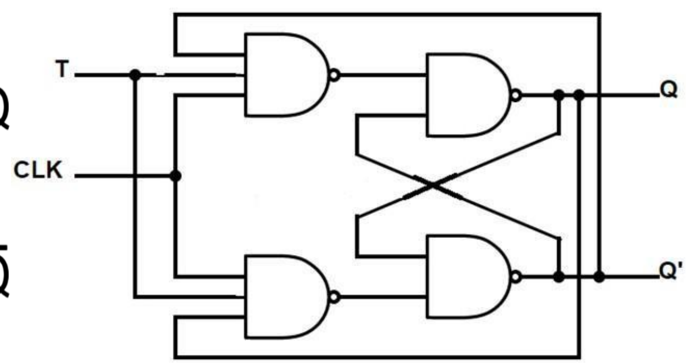
THEORY:

The IC 7476 commonly known as IC 74LS76 is a dual JK flip IC with Set and Clear Input. The Clock, Clear inputs are active low inputs. During the High-Low Clock Transition, input data is transferred to the output. It is also a 16 pin IC. The IC 7476 can be operated from 4.75 to 5.25V and 0 to +70 degrees centigrade temperature. It works with standard TTL voltage and provides a very fast switching speed.

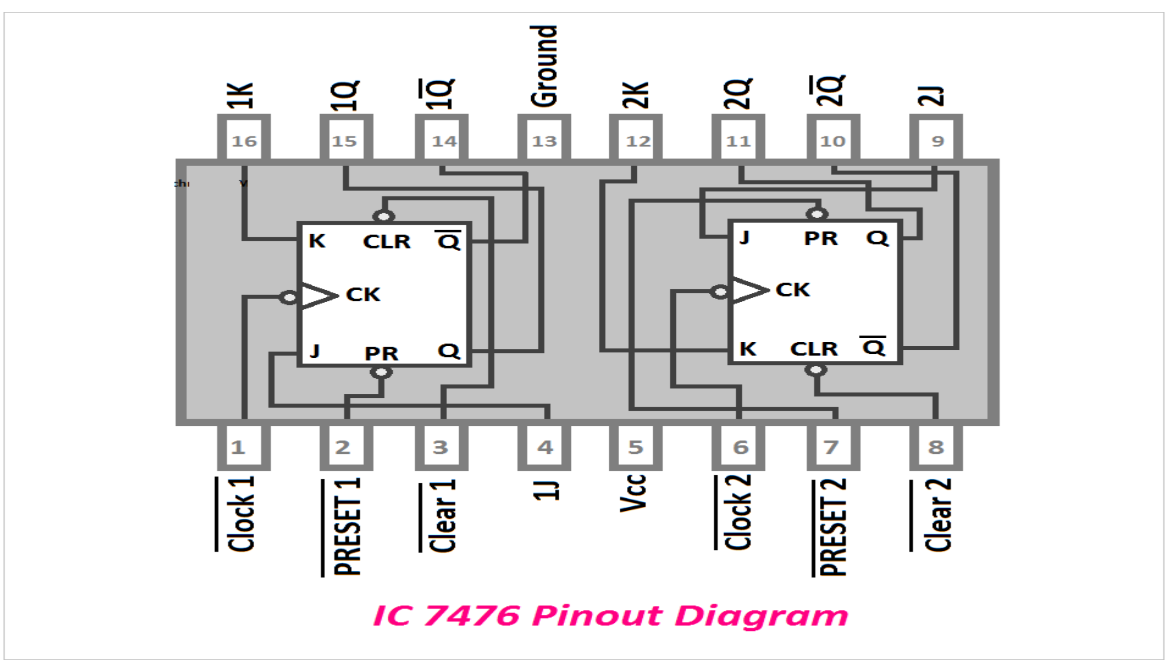
.



JK FLIP-FLOP



T FLIP-FLOP



IC 7476 truth table

The truth table of IC 7476 is given below

INPUTS					OUTPUTS	
PR	CLR	CLK	J	K	Q	$\bar{Q}$
L	H	X	X	X	H	L
H	L	X	X	X	L	H
L	L	X	X	X	H	H
H	H	↑	L	L	Q_0	$\bar{Q}_0$
H	H	↑	H	L	H	L
H	H	↑	L	H	L	H
H	H	↑	H	H	Toggle	

H-High logic level, X-Either LOW or HIGH logic level, L-LOW logic level, ↑-Positive going transition of the clock.

Truth Table of T Flip Flop

CLK	T	Qn	Qn+1
	0	0	0
	0	1	1
	1	0	1
	1	1	0

PROCEDURE:

i. Connections are given as per circuit diagram.
ii. Logical inputs are given as per circuit diagram.
iii. Observe the output and verify the truth table.

RESULT:

Thus the T Flip flop was designed and their truth table is verified.

PRECATIONS:

- All connections should be made neat and tight.
- Digital lab kits and ICs should be handled with utmost care.
- While making connections main voltage should be kept switched off.
- Never touch live and naked wires.

EXPERIMENT – 11

LEARNING OUTCOMES: -

By the end of Lab. Session, the students should be able to construct a decoder (n input lines into 2^n output lines) and encoder (2^n input lines into n output lines) using only basic gates (AND, OR, NOT).

(a) To design 3:8 decoder using logic gate (3 bit binary number to octal number)

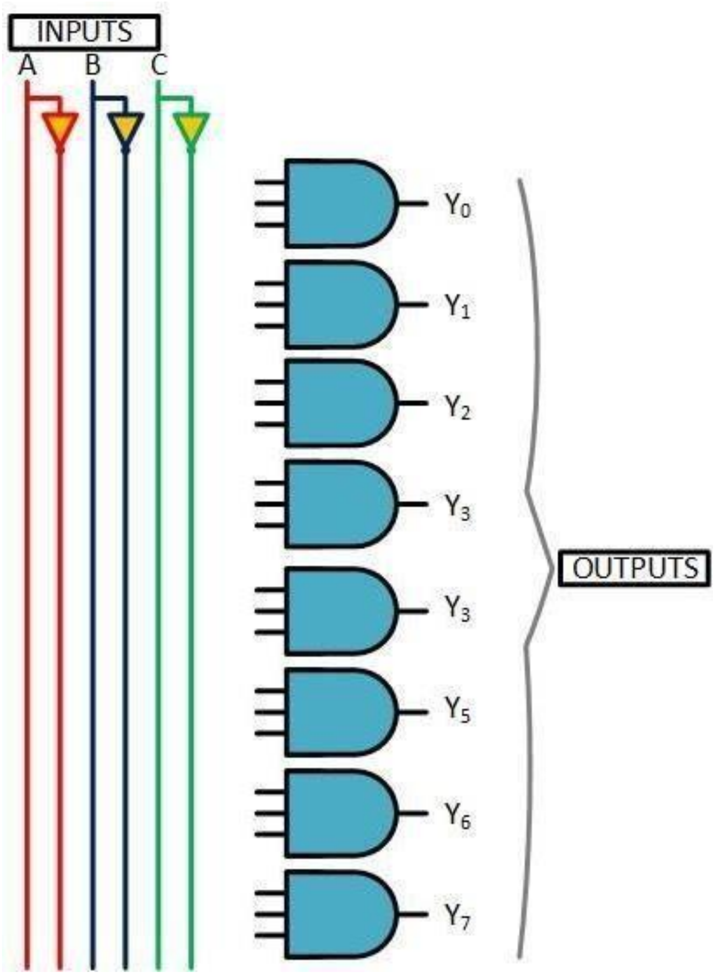
Apparatus

Hardware implementation: Digital trainer kit or IC:- NOT, AND (3input), Bread board, Connecting Wires, Power supply

Theory

A decoder is a combinational logic circuit which is used to change the code into a set of signals. It is the reverse process of an encoder. A decoder circuit takes multiple inputs and gives multiple outputs. A decoder circuit takes binary data of 'n' inputs into '2^n' unique output. In addition to input pins, the decoder has a enable pin. This enables the pin when negated, makes the circuit inactive. In this article, we discuss 3 to 8 line Decoder.

Combinational Logic Circuit:



Truth Table:

Octal Number	Inputs			Outputs							
	A	B	C	Y_0	Y_1	Y_2	Y_3	Y_4	Y_5	Y_6	Y_7
0	0	0	0	1	0	0	0	0	0	0	0
1	0	0	1	0	1	0	0	0	0	0	0
2	0	1	0	0	0	1	0	0	0	0	0
3	0	1	1	0	0	0	1	0	0	0	0
4	1	0	0	0	0	0	0	1	0	0	0
5	1	0	1	0	0	0	0	0	1	0	0
6	1	1	0	0	0	0	0	0	0	1	0
7	1	1	1	0	0	0	0	0	0	0	1

Truth Table 3:8 decoder

PROCEDURE:

1) Connections are given as per circuit diagram.
2) Logical inputs are given as per circuit diagram.
3) Observe the output and verify the truth table.

RESULT:

Thus the DECODER was designed and their truth table is verified.

PRECATIONS:

- All connections should be made neat and tight.
- Digital lab kits and ICs should be handled with utmost care.
- While making connections main voltage should be kept switched off.
- Never touch live and naked wires.

(b) To design ENCODER using logic gate

Apparatus

Hardware implementation: Digital trainer kit or IC:- OR, Bread board, Connecting Wires, Power supply.

Theory

An ***encoder*** is a combinational network that performs the reverse operation of the decoder. An encoder has 2^n or less numbers of inputs and n output lines. The output lines of an encoder generate the binary code for the 2^n input variables. Figure illustrates an eight inputs/three outputs encoder. It may also be referred to as an octal-to-binary encoder where binary codes are generated at outputs according to the input conditions.

An encoder accepts digit on its inputs, such as a decimal or octal digit, and converts it to a coded output, such as a binary or BCD. Encoder can also be devised to encode various symbol and alphabetic characters. This process of converting from familiar symbols or numbers to a coded format is called encoding.

4 to 2 Encoder

Let 4 to 2 Encoder has four inputs Y_3, Y_2, Y_1 & Y_0 and two outputs A_1 & A_0. The block diagram of 4 to 2 Encoder is shown in the following figure.

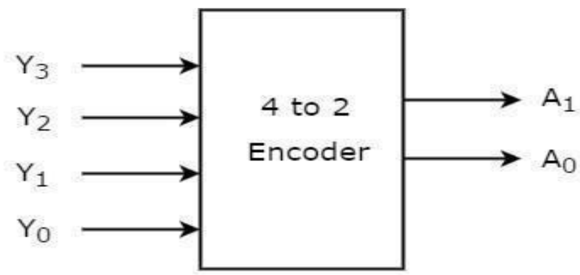


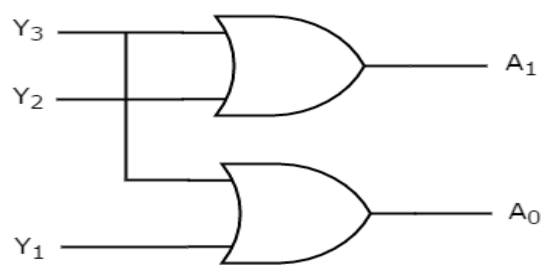
Figure- 4 to 2 Encoder

At any time, only one of these 4 inputs can be '1' in order to get the respective binary code at the output. The Truth table of 4 to 2 encoder is shown below.

Inputs				Outputs	
Y_3	Y_2	Y_1	Y_0	A_1	A_0
0	0	0	1	0	0
0	0	1	0	0	1
0	1	0	0	1	0
1	0	0	0	1	1

The circuit diagram of 4 to 2 Encoder contains two OR gates. These OR gates encode the four inputs with two bits.

Combinational Logic Circuit:



PROCEDURE:

1) Connections are given as per circuit diagram.
2) Logical inputs are given as per circuit diagram.
3) Observe the output and verify the truth table.

RESULT:

Thus the ENCODER was designed and their truth table is verified.

PRECATIONS:

- All connections should be made neat and tight.
- Digital lab kits and ICs should be handled with utmost care.
- While making connections main voltage should be kept switched off.
- Never touch live and naked wires.

EXPERIMENT – 12

LEARNING OUTCOMES: -

By the end of Lab. Session, the students should be able to derive and design the logic circuit for a Mod-8 Up-Down counter (Ripple) using JK or T flip-flops.

Counter is basically used to count the number of clock pulses applied to a flip-flop. It can also be used for Frequency divider, time measurement, frequency measurement, distance measurement, and also for generating square waveforms. In this, the flip-flops are asynchronous counters and are supplied with different clock signals, there may be a delay in producing output. Also, a few numbers of logic gates are needed to design asynchronous counters. So they are elementary in design and also are less expensive.

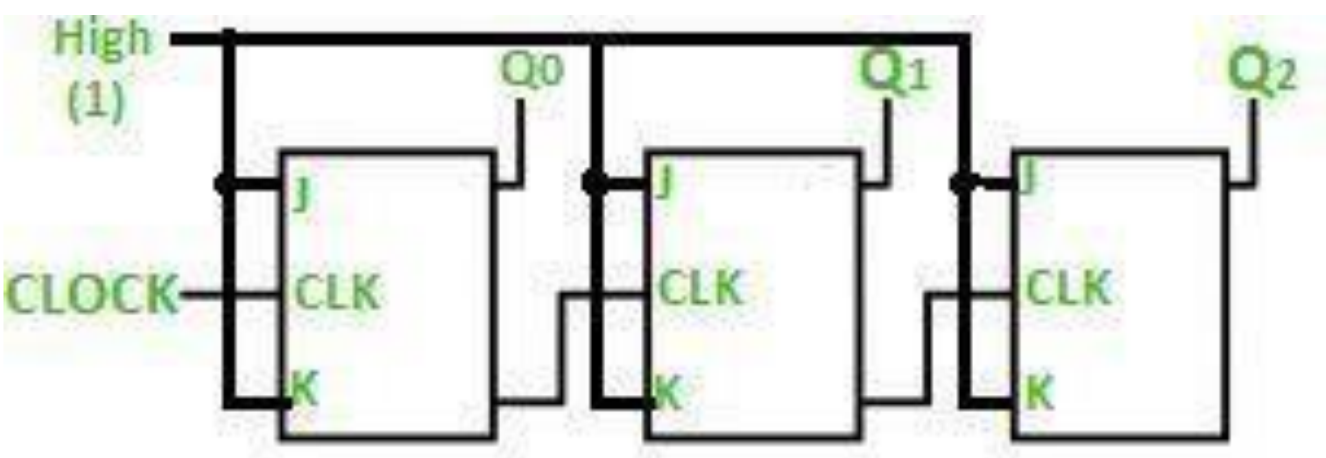
Ripple counter

Ripple counter is a cascaded arrangement of flip-flops where the output of one flip-flop drives the clock input of the following flip-flop. The number of flip flops in the cascaded arrangement depends upon the number of different logic states that it goes through before it repeats the sequence a parameter known as the modulus of the counter. A n-bit ripple counter can count up to 2n states. It is also known as MOD n counter. It is known as ripple counter because of the way the clock pulse ripples its way through the flip-flops. Some of the features of ripple counter are:

- It is an asynchronous counter.
- Different flip-flops are used with a different clock pulse.
- All the flip-flops are used in toggle mode.
- Only one flip-flop is applied with an external clock pulse and another flip-flop clock is obtained from the output of the previous flip-flop.
- The flip-flop applied with an external clock pulse act as LSB (Least Significant Bit) in the counting sequence.

A counter may be an up counter that counts upwards or can be a down counter that counts downwards or can do both i.e. count up as well as count downwards depending on the input control. The sequence of counting usually gets repeated after a limit. When counting up, for the n-bit counter the count sequence goes from 000, 001, 010, ... 110, 111, 000, 001, ... etc. When counting down the count sequence goes in the opposite manner: 111, 110, ... 010, 001, 000, 111, 110, ... etc.

A 3-bit Ripple counter using a JK flip-flop is as follows:



In the circuit shown in the above figure, Q_0 (LSB) will toggle for every clock pulse because JK flip-flop works in toggle mode when both J and K are applied 1, 1, or high input. The following counter will toggle when the previous one changes from 1 to 0.

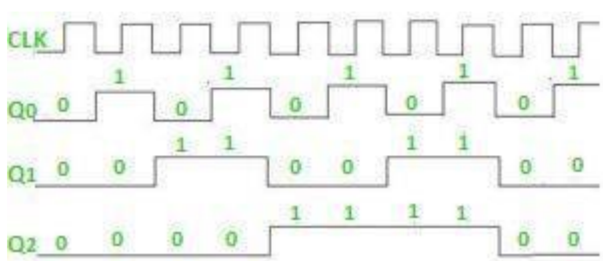
Truth Table is as follows:

Counter State	Q_2	Q_1	Q_0
0	0	0	0
1	0	0	1
2	0	1	0
3	0	1	1
4	1	0	0
5	1	0	1
6	1	1	0
7	1	1	1

The 3-bit ripple counter used in the circuit above has eight different states, each one of which represents a count value. Similarly, a counter having n flip-flops can have a maximum of 2 to the power n states. The number of states that a counter owns is known as its mod (modulo) number. Hence a 3-bit counter is a mod-8 counter. A mod-n counter may also be described as a divide-by-n counter. This is because the most significant flip-flop (the furthest flip-flop from the original clock pulse) produces one pulse for every n pulses at the clock input of the least significant flip-flop (the one triggers by the clock pulse). Thus, the above counter is an example of a divide-by-4 counter.

Timing diagram

Let us assume that the clock is negative edge triggered so the above the counter will act as an up counter because the clock is negative edge triggered and output is taken from Q.



Counters are used very frequently to divide clock frequencies and their uses mainly involve digital clocks and in multiplexing. The widely known example of the counter is parallel to serial data conversion logic.

PROCEDURE:

1) Connections are given as per circuit diagram.
2) Logical inputs are given as per circuit diagram.
3) Observe the output and verify the truth table.

RESULT:

Thus the ENCODER was designed and their truth table is verified.

PRECATIONS:

- All connections should be made neat and tight.
- Digital lab kits and ICs should be handled with utmost care.
- While making connections main voltage should be kept switched off.
- Never touch live and naked wires.

Advantages of Ripple Counter in Digital Logic

- Can be easily designed by T flip-flop or D flip-flop.
- Can be used in low speed circuits & divide by n-counters.
- Used as Truncated counters to design any mode number counters (i.e. Mod 4, Mod 3)

Disadvantages of Ripple Counter in Digital Logic

- Extra flip-flop are needed to do resynchronization.
- To count the sequence of truncated counters, additional feedback logic is needed.
- Propagation delay of asynchronous counters is very large, while counting the large number of bits.
- Counting errors may occur due to propagation delay for high clock frequencies.

Note:

Ripple counters are valuable in digital logic design, allowing for straightforward frequency division and sequence generation. They are easily constructed using T or D flip-flops and have applications in various scenarios, despite their limitations such as propagation delay and the need for additional logic for truncated counting. Understanding the principles of ripple counters is essential for digital circuit designers.

<u>**EXPERIMENT – 13**</u>

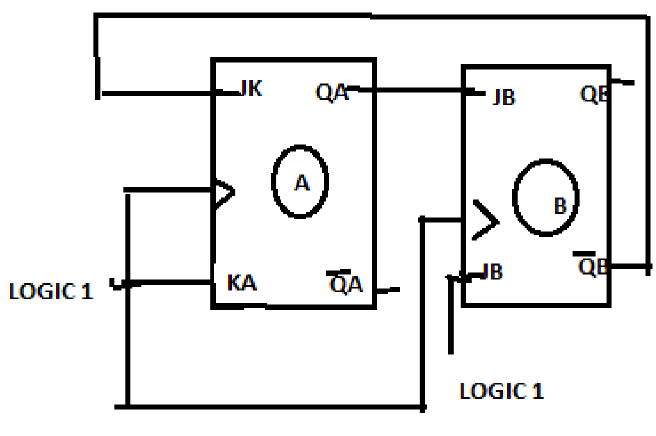
LEARNING OUTCOMES: -

By the end of Lab. Session, the students learn how to design synchronous counters (Mod-3 and Mod-2) using flip-flops (typically JK).

MOD-3 Counter (Synchronous)
A 3-bit synchronous counter, specifically a mod-3 counter, can be implemented using flip-flops and logic gates. It's a digital circuit that counts from 0 to 2 (binary 00 to 10) and then resets to 0, repeating this cycle. This is achieved by using a common clock signal for all flip-flops and carefully connecting their outputs to create the desired counting sequence.
Here's a breakdown of the concept and implementation:
Understanding the Basics:

- **Synchronous Counter:**

  In a synchronous counter, all flip-flops are clocked simultaneously by the same clock signal, ensuring a fixed time relationship between the flip-flop outputs.



- **Modulus (MOD):**

  A MOD-3 counter has 3 states (0, 1, and 2) and resets to 0 after reaching the highest state.

- **Flip-Flops:**
  Typically, J-K or T flip-flops are used. A J-K flip-flop toggles its output when both J and K inputs are high, while a T flip-flop toggles when the T input is high.

- **Applications:**
  Synchronous counters are used in various digital systems, including frequency division, timing circuits, and digital control systems.

  Implementing a 3-bit MOD-3 Synchronous Counter:

- **Truth Table:**
  A truth table can be constructed to show the state transitions of the counter.

CLK count	B (MSB)	A (LSB)	J_A	K_A	J_B	K_B
1	0	0	1	1	0	1
2	0	1	1	1	1	1
3	1	0	0	1	0	1
4	0	0				

PROCEDURE:

1) Connections are given as per circuit diagram.
2) Logical inputs are given as per circuit diagram.
3) Observe the output and verify the truth table.

RESULT:

Thus the MOD-3 Counter was designed and their truth table is verified.

PRECATIONS:

- All connections should be made neat and tight.
- Digital lab kits and ICs should be handled with utmost care.
- While making connections main voltage should be kept switched off.
- Never touch live and naked wires.

MOD-2 counter

A mod-2 asynchronous counter, also known as a divide-by-2 counter, is the simplest form of an asynchronous counter and can be implemented using a single flip-flop. It toggles its output state with each clock pulse, effectively halving the input frequency.

- **Modulus (MOD):**

  A mod-2 counter has a modulus of 2, meaning it has two distinct states (0 and 1).

- **Asynchronous:**

  In an asynchronous counter, the flip-flops are not triggered by the same clock signal directly. The output of one flip-flop acts as the clock for the next flip-flop, creating a ripple effect.

- **Divide-by-2:**
  Because the output toggles once for every two clock cycles, the output frequency is half the input frequency, hence the term "divide-by-2" counter, according to Semiconductor for You.
  Key Characteristics:

How it works:

- **Flip-flop:**

  A JK flip-flop (or a D flip-flop with feedback) is used. In a JK flip-flop, setting both J and K inputs to logic '1' forces the flip-flop to toggle its output state (Q) with each clock pulse.

- **Clock Input:**

  The external clock signal is connected to the clock input of the flip-flop.

- **Output:**

  The output of the flip-flop (Q) toggles between 0 and 1 with each rising or falling edge of the clock signal, depending on the flip-flop's trigger type (positive or negative edge-triggered).

Experiments beyond the curriculum

EXPERIMENT – 14

LEARNING OUTCOMES: -

By the end of Lab. Session, the students gain hands-on experience in programming Arduino using C/C++ or Arduino IDE in the form of Mini projects using programmable ARDUINO.

Introduction: The Arduino Uno is an open-source microcontroller board based on the Microchip ATmega328P microcontroller and developed by Arduino.cc. The board is equipped with sets of digital and analog input/output (I/O) pins that may be interfaced to various expansion boards (shields) and other circuits. The board has 14 digital I/O pins (six capable of PWM output), 6 analog I/O pins, and is programmable with the Arduino IDE (Integrated Development Environment), via a type B USB cable. It can be powered by the USB cable or by an external 9-volt battery, though it accepts voltages between 7 and 20 volts. The word "uno" means "one" in Italian and was chosen to mark the initial release of Arduino Software.

Features of the Arduino

1. Arduino boards are able to read analog or digital input signals from different sensors and turn it into an output such as activating a motor, turning LED on/off, connect to the cloud and many other actions.

2. The board functions can be controlled by sending a set of instructions to the microcontroller on the board via Arduino IDE.
3. Arduino IDE uses a simplified version of C++, making it easier to learn to program.
4. Arduino provides a standard form factor that breaks the functions of the micro- controller into a more accessible package.

Arduino IDE (Integrated Development Environment)
The Arduino Software (IDE) is easy-to-use and is based on the Processing programming environment. The Arduino Integrated Development Environment (IDE) is a cross-platform application (for Windows, macOS, Linux) that is written in functions from C and C++. The open-source Arduino Software (IDE) makes it easy to write code and upload it to the board. This software can be used with any Arduino board.

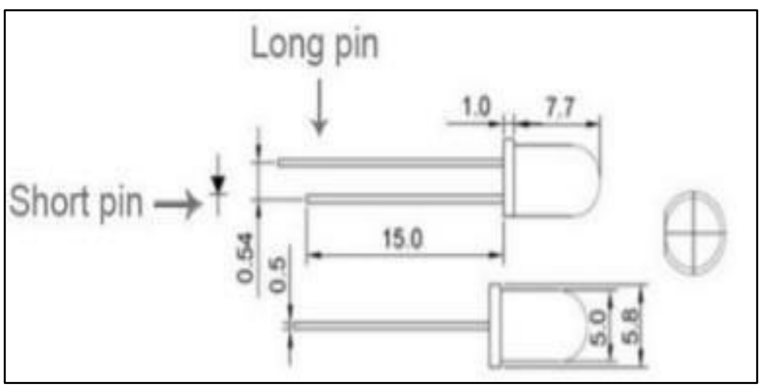
The Arduino Software (IDE) – contains:

- A text editor for writing code
- A message area
- A text consoles
- A toolbar with buttons for common functions and a series of menus. It connects to the Arduino hardware to upload programs and communicate with them.

Installation of Arduino Software (IDE)
To install the Arduino software, download this page: http://arduino.cc/en/Main/Software and proceed with the installation by allowing the driver installation process.

Possible Projects:
Project 1: Controlling the Light Emitting Diode (LED) with a push button.
Project 2: Interfacing of temperature sensor LM35 with Arduino

Project 1
Controlling the Light Emitting Diode (LED) with a push button.
Introduction: Push-button is a very simple mechanism which is used to control electronic signal either by blocking it or allowing it to pass. This happens when mechanical pressure is applied to connect two points of the switch together. Push buttons or switches connect two points in a circuit when pressed. When the push-button is released, there is no connection between the two legs of the push-button. Here it turns on the built-in LED on pin 11 when the button is pressed. The LED stays ON as long as the button is being pressed.
LED Specifications

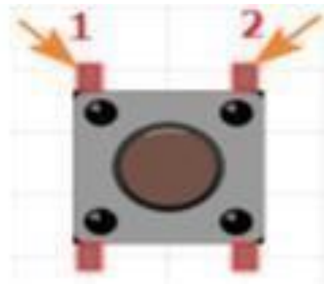


Pin definition

Long pin	+5V
Short pin	GND

Push Button

Specifications:



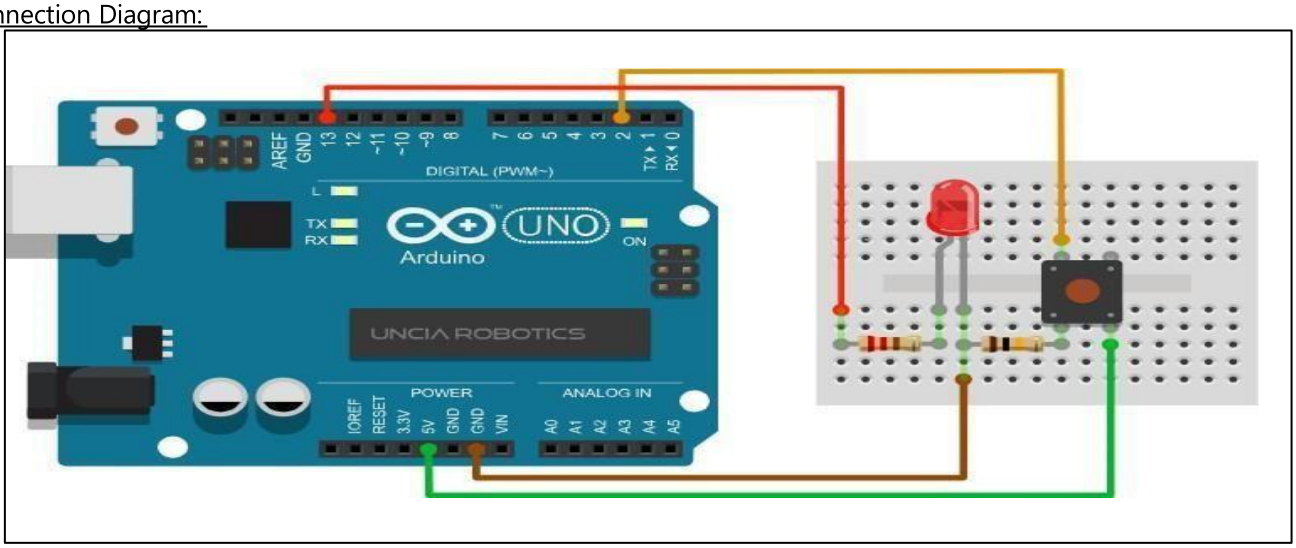
Size	6 x 6 x 5mm
Temperature	-30 ~ +70 Centigrade

Hardware Required:

Component Name	Quantity
Arduino UNO	1
LED	1
Push Button	1
220Ω resistor	1
10KΩ resistor	1
USB Cable	1

Breadboard	1
Jumper wires	Several

Connection Diagram:



Steps of working

1. Insert the push button into your breadboard and connect it to the digital pin 7(D7) which act as INPUT.
2. Insert the LED into the breadboard. Attach the positive leg (the longer leg) to digital pin 11 of the Arduino Uno, and the negative leg via the 220-ohm resistor to GND. The pin D11 is taken as OUTPUT.
3. The 10kΩ resistor used as PULL-UP resistor and 220 Ω resistors is used to limit the current through the LED.
4. Upload the code as given below.
5. Press the push-button to control the ON state of LED.

The Sketch

- This sketch works by setting pin D7 as for the push button as INPUT and pin 11 as an OUTPUT to power the LED.
- The initial state of the button is set to OFF.
- After that the run a loop that continually reads the state from the pushbutton and sends that value as voltage to the LED. The LED will be ON accordingly.

```
/****************Pressing Button LED*****/
const int buttonPin = 7; // choose the pin for the pushbutton
const int led Pin = 11; // choose the pin for a LED
int buttonState = 0;        // variable for reading the pushbutton pin status
void setup()
{
pinMode(ledPin, OUTPUT); // declare LED as output
pinMode(buttonPin, INPUT);        // declare pushbutton as input
}
void loop()
{
buttonState = digitalRead(button Pin); // read input value
if (buttonState == HIGH)
{            // check if the input is HIGH (button pressed)
digitalWrite(ledPin, HIGH); // turn LED ON
}
else
{
digitalWrite(ledPin, LOW); // turn LED OFF}}
```

Observation Table:

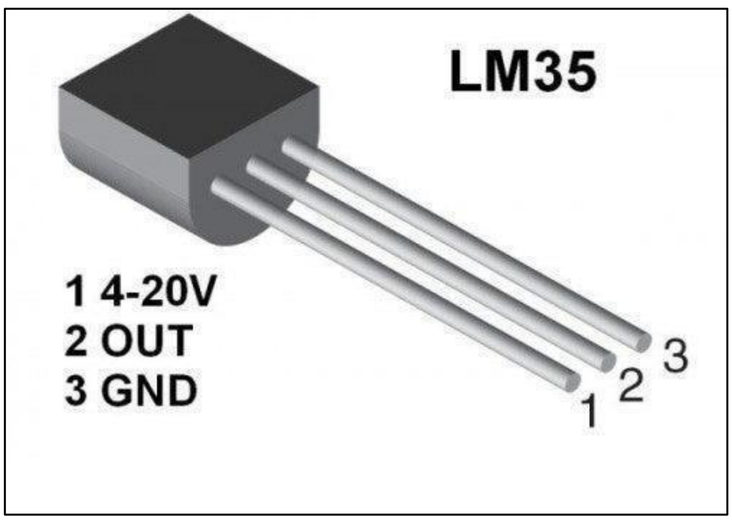
Sr no.	Push button State	LED State
1		
2		

Precautions:

1. The pushbutton is square so it is important to set it appropriately on breadboard.
2. While making the connections make sure to use a pull-down resistor because directly connecting two points of a switch to the circuit will leave the input pin in floating condition and circuit may not work according to the program.
3. It is very important to set pinMode() as OUTPUT first before using digitalWrite() function on that pin.
4. If you do not set the pinMode() to OUTPUT, and connect an LED to a pin, when calling digitalWrite(HIGH), the LED may appear dim.

Project 2
Interfacing of temperature sensor LM35 with Arduino.

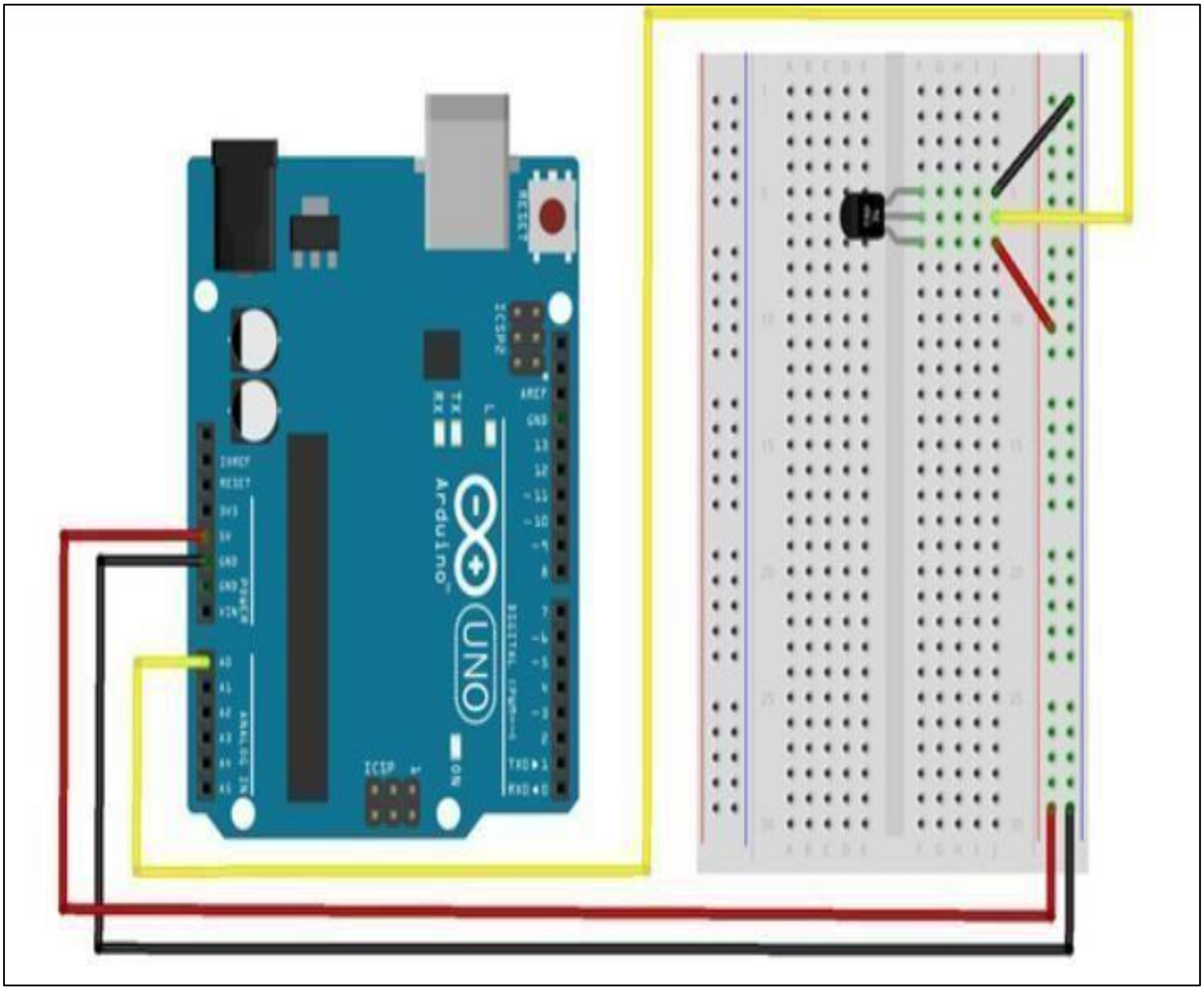
Introduction: The LM35 series are precision integrated-circuit temperature devices with an output voltage linearly proportional to the Centigrade temperature. LM35 is three terminal linear temperature sensors from National semiconductors. It can measure temperature from -55 degree Celsius to +150 degree Celsius. The voltage output of the LM35 increases 10mV per degree Celsius rise in temperature. LM35 can be operated from a 5V supply and the stand by current is less than 60uA. The pin out of LM35 is shown in the figure below.



Hardware Required:

Component Name	Quantity
Arduino UNO	1
Lm35	1
USB Cable	1
Breadboard	1
Jumper wires	Several

Connection Diagram:



Steps of working

1. Insert the temperature sensor into your breadboard and connect its pin1 to the supply.
2. Connect its center pin to the analog pin A0 and the remaining pin3 to GND on the breadboard.
3. Upload the code as given below.
4. Vary the temperature and read the voltage changes.
5. Open the Arduino IDE's serial monitor to see the results.

The Sketch

This sketch works by setting pin A0 as for the temperature sensor. After that the run a loop that continually reads the value from the sensor and sends that value as voltage. The voltage value is between 0–5 volts, when temperature will vary accordingly.

```
/*************File name: LM 35 Temperature Sensor.ino Description: Lit
LM35 Temperature Sensor, let Precision Temperature sensor***/
int LM35Pin=A0;
void setup()
{
Serial.begin(9600);}

void loop ()
{int val;
int data;
val = analogRead(LM35Pin);
data= (val*5)/10;
Serial.print("Temp:");
Serial.print(data);
Serial.println("C");
delay(500);
}
```

Observation Table:

Sr. no.	Voltage	Temperature
1		
2		
3		
4		
5		